

Chapter 22

DESIGN VIA OPTIMAL CONTROL TECHNIQUES

22.1 Preview

The previous chapter gave an introduction to MIMO control-system synthesis by showing how SISO methods could sometimes be used in MIMO problems. However, some MIMO problems require a fundamentally MIMO approach. These are the topic of the current chapter. We will emphasize methods based on optimal control theory. There are three reasons for this choice:

- 1 It is relatively easy to understand.
- 2 It has been used in a myriad of applications. (Indeed, the authors have used these methods on approximately 20 industrial applications.)
- 3 It is a valuable precursor to other advanced methods—e.g., Model Predictive Control, which is explained in the next chapter.

The analysis presented in this chapter builds on the results in Chapter 18, where state space design methods were briefly described in the SISO context. We recall, from that chapter, that the two key elements were

- state estimation by an observer
- state-estimate feedback

We will mirror these elements here for the MIMO case.

22.2 State-Estimate Feedback

Consider the following MIMO state space model having m inputs and p outputs.

$$\dot{x}(t) = \mathbf{A}_o x(t) + \mathbf{B}_o u(t) \quad (22.2.1)$$

$$y(t) = \mathbf{C}_o x(t) \quad (22.2.2)$$

where $x(t) \in \mathbb{R}^n$, $u(t) \in \mathbb{R}^m$, and $y(t) \in \mathbb{R}^p$.

By analogy with state-estimate feedback in the SISO case (as in Chapter 7), we seek a matrix $\mathbf{K} \in \mathbb{R}^{m \times n}$ such that $\mathbf{A}_o - \mathbf{B}_o \mathbf{K}$ has its eigenvalues in the LHP and an observer gain $\mathbf{J}$ such that $\mathbf{A}_o - \mathbf{J} \mathbf{C}_o$ has its eigenvalues in the LHP. Further, we will typically require that the closed-loop poles reside in some specified region in the left-half plane. Tools such as MATLAB provide solutions to these problems. We illustrate by a simple example.

Example 22.1. Consider a MIMO plant having the nominal model

$$\mathbf{G}_o(s) = \frac{1}{s(s+1)(s+2)} \begin{bmatrix} 2(s+1) & -0.5s(s+1) \\ s & 2s \end{bmatrix} \quad (22.2.3)$$

Say that the plant has step-type input disturbances in both channels.

Using state-estimate feedback ideas, design a multivariable controller which stabilizes the plant and, at the same time, ensures zero steady-state error for constant references and disturbances.

Solution

We first build state space models $(\mathbf{A}_p, \mathbf{B}_p, \mathbf{C}_p, \mathbf{0})$ and $(\mathbf{A}_d, \mathbf{B}_d, \mathbf{C}_d, \mathbf{0})$ for the plant and for the input disturbances, respectively. We follow the same basic idea as in the SISO case (in section §18.7)—we estimate not only the plant state $x_p(t)$ but also the disturbance vector $d_i(t)$. We then form the control law

$$u(t) = -\mathbf{K}_p \hat{x}(t) - \hat{d}_i(t) + \bar{r}(t) \quad (22.2.4)$$

We recall that the disturbances are observable from the plant output, although they are unreachable from the plant input.

One pair of possible state space models is

$$\dot{x}_p(t) = \mathbf{A}_p x_p(t) + \mathbf{B}_p u(t) \quad y(t) = \mathbf{C}_p x_p(t) \quad (22.2.5)$$

$$\dot{x}_d(t) = \mathbf{A}_d x_d(t) + \mathbf{B}_d u(t) \quad d_i(t) = \mathbf{C}_d x_d(t) \quad (22.2.6)$$

where

$$\mathbf{A}_p = \begin{bmatrix} -3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}; \quad \mathbf{B}_p = \begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & -0.5 \\ 1 & 0 \end{bmatrix}; \quad \mathbf{C}_p = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (22.2.7)$$

and

$$\mathbf{A}_d = \mathbf{0}; \quad \mathbf{B}_d = \mathbf{0}; \quad \mathbf{C}_d = \mathbf{I}_2 \quad (22.2.8)$$

where $\mathbf{I}_2$ is the identity matrix in $\mathbb{R}^{2 \times 2}$.

The augmented state space model, $(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{0})$, is then given by

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_p & \mathbf{B}_p \mathbf{C}_d \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix} = \begin{bmatrix} \mathbf{A}_p & \mathbf{B}_p \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} \mathbf{B}_p \\ \mathbf{B}_d \end{bmatrix} = \begin{bmatrix} \mathbf{B}_p \\ \mathbf{0} \end{bmatrix} \quad \mathbf{C} = [\mathbf{C}_p \quad \mathbf{0}] \quad (22.2.9)$$

leading to a model with six states.

We then compute the observer gain $\mathbf{J}$, choosing the six observer poles located at $-5, -6, -7, -8, -9, -10$. This is done using the MATLAB command `place` for the pair $(\mathbf{A}^T, \mathbf{C}^T)$.

Next we compute the feedback gain $\mathbf{K}$. We note from (22.2.4) that it is equivalent (with $\bar{r}(t) = 0$) to

$$u(t) = -[\mathbf{K}_p \quad \mathbf{C}_d] \begin{bmatrix} \hat{x}_p(t) \\ \hat{x}_d(t) \end{bmatrix} \Rightarrow \mathbf{K} = [\mathbf{K}_p \quad \mathbf{I}_2] \quad (22.2.10)$$

i.e., we need only compute $\mathbf{K}_p$. This is done by using the MATLAB command `place` for the pair $(\mathbf{A}_p, \mathbf{B}_p)$. The poles in this case are chosen at $-1.5 \pm j1.32, -3$, and -5 . Recall that the other two poles correspond to the uncontrollable disturbance states. They are located at the origin.

The design is evaluated by applying step references and input disturbances in both channels, as follows:

$$r_1(t) = \mu(t-2); \quad r_2(t) = -\mu(t-5); \quad d_i^{(1)}(t) = \mu(t-10); \quad d_i^{(2)}(t) = \mu(t-15) \quad (22.2.11)$$

where $d_i^{(1)}(t)$ and $d_i^{(2)}(t)$ are the first and second components of the input-disturbance vector respectively.

The results are shown in Figure 22.1

The design was based on an arbitrary choice of observer and controller poles (polynomials $E(s)$ and $F(s)$, respectively). Thus, the apparent triangular nature of the coupling in the design was not a design objective. However, the choice of the control law ensures that the disturbances are completely compensated in steady state.

The file `mimo2.mdl` contains the SIMULINK schematic with the MIMO control loop for this example.

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We will not develop the idea of pole assignment further. Instead, we turn to an alternative procedure that deals with the MIMO case via optimization methods. A particularly nice approach for the design of $\mathbf{K}$ and $\mathbf{J}$ is to use quadratic optimization, because it leads to simple closed-form solutions. The details are provided in the following sections.

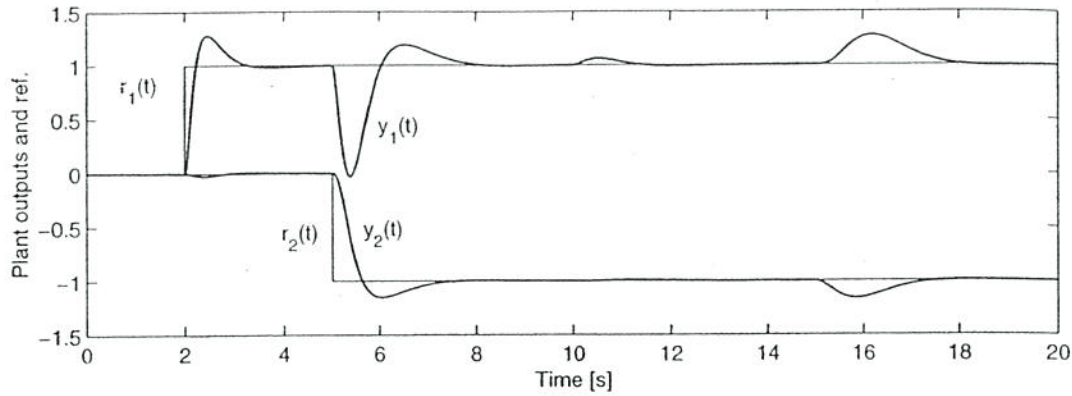


Figure 22.1. MIMO design based in state-estimate feedback

22.3 Dynamic Programming and Optimal Control

Consider a general nonlinear system with input $u(t) \in \mathbb{R}^m$, described in state space form by

$$\frac{dx(t)}{dt} = f(x(t), u(t), t) \quad (22.3.1)$$

where $x(t) \in \mathbb{R}^n$, together with the following optimization problem:

Problem (General optimal control problem). Find an optimal input $u^o(t)$, for $t \in [t_o, t_f]$, such that

$$u^o(t) = \arg \min_{u(t)} \left\{ \int_{t_o}^{t_f} \mathcal{V}(x, u, t) dt + g(x(t_f)) \right\} \quad (22.3.2)$$

where $\mathcal{V}(x, u, t)$ and $g(x(t_f))$ are nonnegative functions.

One possible way to solve this problem is to use Dynamic Programming ; this approach is based on the following result.

Theorem 22.1 (Optimality Principle (Bellman)). If $\{u(t) = u^o(t), t \in [t_o, t_f]\}$ is the optimal solution for the above problem, then $u^o(t)$ is also the optimal solution over the (sub)interval $[t_o + \Delta t, t_f]$, where $t_o < t_o + \Delta t < t_f$.

Proof

(By contradiction)

Denote by $x^o(t)$ the state evolution that results from applying $u^o(t)$ over the whole interval, i.e., for $t \in [t_o, t_f]$. We can then describe the optimal cost over the interval as

$$\int_{t_o}^{t_o+\Delta t} \mathcal{V}(x^o, u^o, t) dt + \int_{t_o+\Delta t}^{t_f} \mathcal{V}(x^o, u^o, t) dt + g(x^o(t_f)) \quad (22.3.3)$$

Assume now that there exists an input $\tilde{u}(t)$ such that

$$\tilde{u}(t) = \arg \min_{u(t)} \left\{ \int_{t_o+\Delta t}^{t_f} \mathcal{V}(x, u, t) dt + g(x(t_f)) \right\} \quad (22.3.4)$$

with corresponding state trajectory $\tilde{x}(t)$ such that $\tilde{x}(t_o + \Delta t) = x^o(t_o + \Delta t)$. Assume that

$$\int_{t_o+\Delta t}^{t_f} \mathcal{V}(\tilde{x}, \tilde{u}, t) dt + g(\tilde{x}(t_f)) < \int_{t_o+\Delta t}^{t_f} \mathcal{V}(x^o, u^o, t) dt + g(x^o(t_f)) \quad (22.3.5)$$

Consider now the policy of applying $u(t) = u^o(t)$ in the interval $[t_o, t_o + \Delta t]$ and then $u(t) = \tilde{u}(t)$ over the interval $[t_o + \Delta t, t_f]$. The resulting cost over the complete interval would be

$$\int_{t_o}^{t_o+\Delta t} \mathcal{V}(x^o, u^o, t) dt + \int_{t_o+\Delta t}^{t_f} \mathcal{V}(\tilde{x}, \tilde{u}, t) dt + g(\tilde{x}(t_f)) \quad (22.3.6)$$

By comparing (22.3.3) and (22.3.6), then, upon using (22.3.5), we conclude that the policy of applying $u(t) = u^o(t)$ for $t \in [t_o, t_o + \Delta t]$ and $\tilde{u}(t)$ for $t \in [t_o + \Delta t, t_f]$ would yield a smaller cost than applying $u(t) = u^o(t)$ over the entire interval. However this contradicts the assumption regarding the optimality of $u^o(t)$ over that interval.

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We will next use Theorem 22.1 to derive necessary conditions for the optimal u . The idea is to consider a general time interval $[t, t_f]$, where $t \in [t_o, t_f]$, and then to use the Optimality Principle with an infinitesimal time interval $[t, t + \Delta t]$.

Denote by $J^o(x(t), t)$ the optimal (minimal) cost in the interval $[t, t_f]$, with initial state $x(t)$:

$$J^o(x(t), t) = \min_{u(\tau)} \left\{ \int_t^{t_f} \mathcal{V}(x, u, \tau) d\tau + g(x(t_f)) \right\} \quad (22.3.7)$$

Then, upon applying Theorem 22.1, we have that

$$J^o(x(t), t) = \min_{\substack{u(\tau) \\ \tau \in [t, t+\Delta t]}} \left\{ \int_t^{t+\Delta t} \mathcal{V}(x, u, \tau) d\tau + J^o(x(t+\Delta t), t+\Delta t) \right\} \quad (22.3.8)$$

We shall now consider small perturbations around the point $(x(t), t)$, considering Δt to be a very small quantity. To achieve this, we expand $J^o(x(t+\Delta t), t+\Delta t)$ in a Taylor series. We thus obtain

$$\begin{aligned} J^o(x(t), t) = \min_{u(t)} & \left\{ \mathcal{V}(x(t), u(t), t) \Delta t + \right. \\ & \left. J^o(x(t), t) + \frac{\partial J^o(x(t), t)}{\partial t} \Delta t + \left[\frac{\partial J^o(x(t), t)}{\partial x} \right]^T (x(t+\Delta t) - x(t)) + \mathcal{O}(x, t) \right\} \end{aligned} \quad (22.3.9)$$

where $\mathcal{O}(x, t)$ includes the high-order terms in the Taylor expansion.

If we now let $\Delta t \rightarrow dt$, then (22.3.9) yields

$$-\frac{\partial J^o(x(t), t)}{\partial t} = \min_{u(t)} \{ \mathcal{W}(x(t), u(t), t) \} \quad (22.3.10)$$

where

$$\mathcal{W}(x(t), u(t), t) = \mathcal{V}(x(t), u(t), t) + \left[\frac{\partial J^o(x(t), t)}{\partial x} \right]^T f(x(t), u(t), t) \quad (22.3.11)$$

Note that we have used the fact that

$$\lim_{\Delta t \rightarrow 0} \frac{x(t+\Delta t) - x(t)}{\Delta t} = \frac{dx(t)}{dt} \quad (22.3.12)$$

and Equation (22.3.1).

The optimal $u(t)$ satisfying the right-hand side of Equation (22.3.10) can be expressed symbolically as

$$u^o(t) = \mathcal{U} \left(\frac{\partial J^o(x(t), t)}{\partial x}, x(t), t \right) \quad (22.3.13)$$

which, when substituted into (22.3.10) leads to the following equations for the optimal cost:

$$-\frac{\partial J^o(x(t), t)}{\partial t} = \mathcal{V}(x(t), \mathcal{U}, t) + \left[\frac{\partial J^o(x(t), t)}{\partial x} \right]^T f(x(t), \mathcal{U}, t) \quad (22.3.14)$$

The solution for this equation must satisfy the boundary condition

$$J^o(x(t_f), t_f) = g(x(t_f)) \quad (22.3.15)$$

At this stage we cannot proceed further without being more specific about the nature of the original problem. We also note that we have implicitly assumed that the function $J^o(x(t), t)$ is well behaved, which means that it is continuous in its arguments and that it can be expanded in a Taylor series. It has been proved elsewhere (see references at the end of the chapter) that these conditions are sufficient to make (22.3.13) the optimal solution.

22.4 The Linear Quadratic Regulator (LQR)

We next apply the above general theory to the following problem.

Problem (The LQR problem). *Consider a linear time-invariant system having a state space model, as defined below:*

$$\frac{dx(t)}{dt} = \mathbf{A}x(t) + \mathbf{B}u(t) \quad x(t_o) = x_o \quad (22.4.1)$$

$$y(t) = \mathbf{C}x(t) + \mathbf{D}u(t) \quad (22.4.2)$$

where $x \in \mathbb{R}^n$ is the state vector, $u \in \mathbb{R}^m$ is the input, $y \in \mathbb{R}^p$ is the output, $x_o \in \mathbb{R}^n$ is the state vector at time $t = t_o$, and $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ are matrices of appropriate dimensions.

Assume that we aim at driving the initial state x_o to the smallest possible value as soon as possible in the interval $[t_o, t_f]$, but without spending too much control effort (as measured by the magnitude of u) to achieve that goal. Then the optimal-regulator problem is defined as the problem of finding an optimal control $u(t)$ over the interval $[t_o, t_f]$ such that the following cost function is minimized:

$$J_u(x(t_o), t_o) = \int_{t_o}^{t_f} [x(t)^T \Psi x(t) + u(t)^T \Phi u(t)] dt + x(t_f)^T \Psi_f x(t_f) \quad (22.4.3)$$

where $\Psi \in \mathbb{R}^{n \times n}$ and $\Psi_f \in \mathbb{R}^{n \times n}$ are symmetric nonnegative definite matrices and $\Phi \in \mathbb{R}^{m \times m}$ is a symmetric positive definite matrix.

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To solve this problem, the theory summarized in section §22.3 can be used. We first make the following connections between the general optimal problem and the LQR problem:

$$f(x(t), u(t), t) = \mathbf{A}x(t) + \mathbf{B}u(t) \quad (22.4.4)$$

$$\mathcal{V}(x, u, t) = x(t)^T \Psi x(t) + u(t)^T \Phi u(t) \quad (22.4.5)$$

$$g(x(t_f)) = x(t_f)^T \Psi_f x(t_f) \quad (22.4.6)$$

Then

$$\begin{aligned} \mathcal{W}(x(t), u(t), t) &\triangleq \mathcal{V}(x(t), u(t), t) + \left[\frac{\partial J^o(x(t), t)}{\partial x} \right]^T f(x(t), u(t), t) \\ &= x(t)^T \Psi x(t) + u(t)^T \Phi u(t) + \left[\frac{\partial J^o(x(t), t)}{\partial x} \right]^T (\mathbf{A}x(t) + \mathbf{B}u(t)) \end{aligned} \quad (22.4.7)$$

Thus, to obtain the optimal u as given by (22.3.13), we have to minimize $\mathcal{W}(x(t), u(t), t)$. This requires us to compute the gradient of $\mathcal{W}(x(t), u(t), t)$ with respect to u and then set this gradient equal to zero.

$$\frac{\partial \mathcal{W}}{\partial u} = 2\Phi u(t) + \mathbf{B}^T \frac{\partial J^o(x(t), t)}{\partial x} \quad (22.4.8)$$

Now, setting $\frac{\partial \mathcal{W}}{\partial u} = 0$ shows that the optimal input, $u^o(t)$, is given by

$$u^o(t) = -\frac{1}{2} \Phi^{-1} \mathbf{B}^T \frac{\partial J^o(x(t), t)}{\partial x} \quad (22.4.9)$$

We also observe that the Hessian of $\mathcal{W}$ with respect to u is equal to Φ , which, by initial assumption, is positive definite. This confirms that (22.4.9) gives a minimum for $\mathcal{W}$.

We next note that $J^o(x^o(t_f), t_f) = [x^o(t_f)]^T \Psi_f x^o(t_f)$. We observe that this is a quadratic function of the state at time t_f . We show by induction that this is true at every time t . We proceed by assuming that the optimal cost has the form

$$J^o(x(t), t) = x^T(t) \mathbf{P}(t) x(t) \quad \text{with} \quad \mathbf{P}(t) = [\mathbf{P}(t)]^T \quad (22.4.10)$$

We then have that

$$\frac{\partial J^o(x(t), t)}{\partial x} = 2\mathbf{P}(t)x(t) \quad (22.4.11)$$

$$\frac{\partial J^o(x(t), t)}{\partial t} = x^T(t) \frac{d\mathbf{P}(t)}{dt} x(t) \quad (22.4.12)$$

If we now use (22.4.12) in (22.4.9),

the optimal control can be expressed as

$$u^o(t) = -\mathbf{K}_u(t)x(t) \quad (22.4.13)$$

where $\mathbf{K}_u(t)$ is a time-varying gain, given by

$$\mathbf{K}_u(t) = \Phi^{-1}\mathbf{B}^T\mathbf{P}(t) \quad (22.4.14)$$

We also have from (22.4.7) that

$$\mathcal{W}(x(t), u^o(t), t) = x^T(t)(\Psi - \mathbf{P}(t)\mathbf{B}\Phi^{-1}\mathbf{B}^T\mathbf{P}(t) + 2\mathbf{P}(t)\mathbf{A})x(t) \quad (22.4.15)$$

To compute $\mathbf{K}_u(t)$, we first need to obtain $\mathbf{P}(t)$, which can be done by using (22.4.12) and (22.4.15) in (22.3.14), leading to

$$-x^T(t)\frac{d\mathbf{P}(t)}{dt}x(t) = x^T(t)(\Psi - \mathbf{P}(t)\mathbf{B}\Phi^{-1}\mathbf{B}^T\mathbf{P}(t) + 2\mathbf{P}(t)\mathbf{A})x(t) \quad (22.4.16)$$

We also note that $2x^T(t)\mathbf{P}(t)\mathbf{A}x(t) = x^T(t)(\mathbf{P}(t)\mathbf{A} + \mathbf{A}^T\mathbf{P}(t))x(t)$. For (22.4.16) to hold for all $x(t)$, we require that

$$-\frac{d\mathbf{P}(t)}{dt} = \Psi - \mathbf{P}(t)\mathbf{B}\Phi^{-1}\mathbf{B}^T\mathbf{P}(t) + \mathbf{P}(t)\mathbf{A} + \mathbf{A}^T\mathbf{P}(t) \quad (22.4.17)$$

Equation (22.4.17) is known as the **Continuous Time Dynamic Riccati Equation (CTDRE)**. This equation has to be solved backwards in time, to satisfy the boundary condition (22.4.6), (22.4.10):

$$\mathbf{P}(t_f) = \Psi_f \quad (22.4.18)$$

The above theory holds equally well for time-varying systems—i.e., when $\mathbf{A}, \mathbf{B}, \Phi, \Psi$ are all functions of time. However, in the time-invariant case, one can say much more about the properties of the solution. This is the subject of the next section.

22.5 Properties of the Linear Quadratic Optimal Regulator

Here we assume that $\mathbf{A}, \mathbf{B}, \Phi, \Psi$ are all time-invariant. We will be particularly interested in what happens at $t \rightarrow \infty$. We will summarize the key results here.

However, the reader is also referred to Appendix D, where a treatment of the results in greater depth is given. We will outline two versions of the results. In §22.5.1 we give a quick review of the basic properties as used in simple problems. In §22.5.2 we will outline more advanced results, such as might be needed in more sophisticated situations.

22.5.1 Quick Review of Properties

In many simple problems, two key assumptions hold:

- (i) The system $(\mathbf{A}, \mathbf{B})$ is stabilizable from $u(t)$. (The reader may feel that this assumption is reasonable—after all if, the system isn't stabilizable, then clearly optimal control will not help recover the situation.)
- (ii) The system states are all adequately *seen* by the cost function (22.4.3). Technically, this is stated as requiring that $(\Psi^{\frac{1}{2}}, \mathbf{A})$ be detectable.

Under these conditions, the solution to the CTDRE, $\mathbf{P}(t)$, converges to a steady-state limit $\mathbf{P}_{\infty}^s$ as $t_f \rightarrow \infty$. This limit has two key properties:

- $\mathbf{P}_{\infty}^s$ is the only nonnegative solution of the matrix algebraic Riccati equation (22.5.1), in §22.5.2 obtained by setting $\frac{d\mathbf{P}(t)}{dt} = 0$ in (22.4.17).
- When this steady-state value is used to generate a feedback control law, as in (22.4.14), then the resulting closed-loop system is *stable*.

Clearly, the last property is of major importance.

22.5.2 More Detailed Review of Properties

This section refers frequently to Appendix D, so the reader is encouraged to view the appendix on his or her computer whilst reading this section.

In some applications, it is useful to know a bit more about the properties of the solution to the Optimal Control Problem than was given in subsection §22.5.1. Further results are outlined here.

Lemma 22.1. *If $\mathbf{P}(t)$ converges as $t_f \rightarrow \infty$, then the limiting value $\mathbf{P}_{\infty}$ satisfies the following Continuous-Time Algebraic Riccati Equation (CTARE):*

$$\mathbf{0} = \Psi - \mathbf{P}_{\infty} \mathbf{B} \Phi^{-1} \mathbf{B}^T \mathbf{P}_{\infty} + \mathbf{P}_{\infty} \mathbf{A} + \mathbf{A}^T \mathbf{P}_{\infty} \quad (22.5.1)$$

Furthermore, provided $(\mathbf{A}, \mathbf{B})$ is stabilizable and $(\mathbf{A}, \Psi^{\frac{1}{2}})$ has no unobservable modes on the imaginary axis, then there exists a unique positive semidefinite solution

$\mathbf{P}_\infty^s$ to the CTARE having the property that the system matrix of the closed-loop system, $\mathbf{A} - \Phi^{-1}\mathbf{B}^T\mathbf{P}_\infty^s$, has all its eigenvalues in the OLHP. We call this solution the stabilizing solution of the CTARE. Other properties of the stabilizing solution are as follows:

- a) If $(\mathbf{A}, \Psi^{\frac{1}{2}})$ is detectable, the stabilizing solution is the only nonnegative solution of the CTARE.
- b) If $(\mathbf{A}, \Psi^{\frac{1}{2}})$ has unobservable modes in the OLHP, then the stabilizing solution is not positive definite.
- c) If $(\mathbf{A}, \Psi^{\frac{1}{2}})$ has an unobservable pole outside the OLHP, then, in addition to the stabilizing solution, there exists at least one other nonnegative solution to the CTARE. However, in this case, the stabilizing solution satisfies $\mathbf{P}_\infty^s - \bar{\mathbf{P}}_\infty \geq 0$, where $\bar{\mathbf{P}}_\infty$ is any other solution of the CTARE.

Proof

See Appendix D.

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Convergence of the solution of the CTDRE to the stabilizing solution of the CTARE is addressed in the following lemma.

Lemma 22.2. *Provided that $(\mathbf{A}, \mathbf{B})$ is stabilizable, that $(\mathbf{A}, \Psi^{\frac{1}{2}})$ has no unobservable poles on the imaginary axis, and that $\Psi > \mathbf{P}_\infty^s$, then*

$$\lim_{t_f \rightarrow \infty} \mathbf{P}(t) = \mathbf{P}_\infty^s \quad (22.5.2)$$

(Provided that $(\mathbf{A}, \Psi^{\frac{1}{2}})$ is detectable, then $\Psi \geq 0$ suffices.)

Proof

See Appendix D.

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We illustrate the above properties by a simple example.

Example 22.2. *Consider the scalar system*

$$\dot{x}(t) = ax(t) + u(t) \quad (22.5.3)$$

and the cost function

$$J = \psi_f x(t_f)^2 + \int_0^{t_f} (\psi x(t)^2 + u(t)^2) dt \quad (22.5.4)$$

22.2.1 Discuss this optimal control problem in the light of Lemma 22.1 and Lemma 22.2.

22.2.2 Discuss the convergence of the solutions of the CTDRE to P_∞^s .

Solution

22.2.1 The associated CTDRE is

$$\dot{P}(t) = -2aP(t) + P(t)^2 - \psi; \quad P(t_f) = \psi_f \quad (22.5.5)$$

and the CTARE is

$$(P_\infty^s)^2 - 2aP_\infty^s - \psi = 0 \quad (22.5.6)$$

Case 1 $\psi \neq 0$

Here, $(A, \Psi^{\frac{1}{2}})$ is completely observable (and thus detectable). From Lemma 22.1, part (a), there is only one nonnegative solution of the CTARE. This solution coincides with the stabilizing solution. Making the calculations, we find that the only nonnegative solution of the CTARE is

$$P_\infty^s = \frac{2a + \sqrt{4a^2 + 4\psi}}{2} \quad (22.5.7)$$

leading to the following feedback gain:

$$K_\infty^s = a + \sqrt{a^2 + \psi} \quad (22.5.8)$$

The corresponding closed-loop pole is at

$$p_{cl} = -\sqrt{a^2 + \psi} \quad (22.5.9)$$

This is clearly in the LHP, verifying that the solution is indeed the stabilizing solution.

Case 2 $\psi = 0$

Here, we will consider three positive values for a : one positive, one negative, and one zero.

(i) $a > 0$. In this case, $(A, \Psi^{\frac{1}{2}})$ has an observable pole outside the stability region. Then, from part (c) of Lemma 22.1, in addition to the stabilizing solution, there exists at least one other nonnegative solution to the CTARE. Making the calculations, we find that there

are two nonnegative solutions of the CTARE: the stabilizing solution $\mathbf{P}_\infty^s = 2a$, and one other solution $\mathbf{P}'_\infty$.

We see that $\mathbf{P}_\infty^s - \mathbf{P}'_\infty > 0$ and that $\mathbf{P}_\infty^s$ gives a feedback-loop gain of $\mathbf{K}_\infty^s = 2a$, leading to closed-loop poles at $p_{cl} = -a$. This is clearly in the LHP, verifying that $\mathbf{P}_\infty^s$ is indeed the stabilizing solution.

- (ii) $a < 0$. Here, $(\mathbf{A}, \Psi^{\frac{1}{2}})$ is again detectable. Thus, from Lemma 22.1 on page 688, part (a), there is only one nonnegative solution of the CTARE. This solution coincides with the stabilizing solution. Making the calculations, we find $\mathbf{P}_\infty^s = 0$, and the corresponding feedback gain is $\mathbf{K}_\infty^s = 0$, leading to closed-loop poles at $p_{cl} = a$, which is stable, because $a < 0$, by hypothesis.
- (iii) $a = 0$. Here, $(\mathbf{A}, \Psi^{\frac{1}{2}})$ has an unobservable pole on the stability boundary. This is, then, not covered by Lemma 22.1 on page 688. However, it can be verified that the CTARE has only one solution, namely $\mathbf{P}_\infty$, which is not stabilizing.

22.2.2 To study the convergence of the solutions, we again consider two cases.

Case 1 $\psi \neq 0$

Here $(\mathbf{A}, \Psi^{\frac{1}{2}})$ is completely observable. Then $\mathbf{P}(t)$ converges to $\mathbf{P}_\infty^s$ given in (22.5.7) for any $\mathbf{P}(t_f) \geq 0$.

Case 2 $\psi = 0$

- (i) $a > 0$. Here, $(\mathbf{A}, \Psi^{\frac{1}{2}})$ has no unobservable poles on the $j\omega$ axis, but it is not detectable. Thus, $\mathbf{P}(t)$ converges to $\mathbf{P}_\infty^s$, provided that we choose $\mathbf{P}(t_f) > \mathbf{P}_\infty^s$.
- (ii) $a < 0$. Here, $(\mathbf{A}, \Psi^{\frac{1}{2}})$ is again detectable, and $\mathbf{P}(t)$ converges to $\mathbf{P}_\infty^s$ for any $\mathbf{P}(t_f) \geq 0$.
- (iii) $a = 0$. Here $(\mathbf{A}, \Psi^{\frac{1}{2}})$ has an unobservable pole on the $j\omega$ axis. Thus, Lemma 22.2 on page 689 does not apply. Actually, $\mathbf{P}(t)$ converges to $\mathbf{0}$ for any $\mathbf{P}(t_f) \geq 0$. However, in this case, zero is not a stabilizing solution.

□□□

Linear quadratic regulator theory is a powerful tool in control-system design. We illustrate its versatility in the next section by using it to solve the so-called Model Matching Problem (MMP).

22.6 Model Matching Based on Linear Quadratic Optimal Regulators

22.6.1 Problem Formulation

Many problems in control synthesis can be reduced to a problem of the following type:

Given two stable transfer functions $M(s)$ and $N(s)$, find a stable transfer function $\Gamma(s)$ so that $N(s)\Gamma(s)$ is close to $M(s)$ in a quadratic norm sense.

We call this a model-matching problem. When $M(s)$ and $N(s)$ are matrix transfer functions, we need to define a suitable norm to measure closeness. By way of illustration, we consider a matrix $A = [a_{ij}] \in \mathbb{C}^{p \times m}$ for which we define the *Fröbenius norm* as follows

$$\|A\|_F = \sqrt{\text{trace } A^H A} = \sqrt{\sum_{i=1}^p \sum_{j=1}^m |a_{ij}|^2} \quad (22.6.1)$$

Under this norm, a suitable synthesis criterion might be

$$\Gamma^o = \arg \min_{\Gamma \in \mathcal{S}} J_\Gamma \quad (22.6.2)$$

where

$$J_\Gamma = \frac{1}{2\pi} \int_{-\infty}^{\infty} \|M(j\omega) - N(j\omega)\Gamma(j\omega)\|_F^2 d\omega \quad (22.6.3)$$

and $\mathcal{S}$ is the class of stable transfer functions.

This problem can be converted into vector form by vectorizing M and Γ . For example, say that Γ is constrained to be lower triangular and that M , N , and Γ are 3×2 , 3×2 , and 2×2 matrices, respectively; then we can write

$$J_\Theta = \frac{1}{2\pi} \int_{-\infty}^{\infty} \|V(j\omega) - W(j\omega)\Theta(j\omega)\|_2^2 d\omega \quad (22.6.4)$$

where $\|\cdot\|_2$ denotes the usual Euclidean vector norm and where, in this special case,

$$\mathbf{V}(s) = \begin{bmatrix} M_{11}(s) \\ M_{12}(s) \\ M_{21}(s) \\ M_{22}(s) \\ M_{31}(s) \\ M_{32}(s) \end{bmatrix}; \quad \mathbf{W}(s) = \begin{bmatrix} N_{11}(s) & N_{12}(s) & 0 \\ 0 & 0 & N_{12}(s) \\ N_{21}(s) & N_{22}(s) & 0 \\ 0 & 0 & N_{22}(s) \\ N_{31}(s) & N_{32}(s) & 0 \\ 0 & 0 & N_{32}(s) \end{bmatrix}; \quad \Theta(s) = \begin{bmatrix} \Gamma_{11}(s) \\ \Gamma_{21}(s) \\ \Gamma_{22}(s) \end{bmatrix} \quad (22.6.5)$$

22.6.2 Conversion to Time Domain

We next select a state space model for $\mathbf{V}(s)$ and $\mathbf{W}(s)$ of the form

$$\mathbf{V}(s) = \mathbf{C}_1[s\mathbf{I} - \mathbf{A}_1]^{-1}\mathbf{B}_1 \quad (22.6.6)$$

$$\mathbf{W}(s) = \mathbf{C}_2[s\mathbf{I} - \mathbf{A}_2]^{-1}\mathbf{B}_2 \quad (22.6.7)$$

where, in the special case of equation (22.6.5), $\mathbf{A}_1$, $\mathbf{B}_1$, $\mathbf{C}_1$, $\mathbf{A}_2$, $\mathbf{B}_2$, and $\mathbf{C}_2$ are matrices of dimensions $n_1 \times n_1$, $n_1 \times 1$, $6 \times n_1$, $n_2 \times n_2$, $n_2 \times 3$, and $6 \times n_2$, respectively.

Before proceeding to solve the model-matching problem, we make a slight generalization. In particular, it is sometimes desirable to restrict the size of Θ . We do this by generalizing (22.6.4) by introducing an extra term that weights Θ . This leads to

$$J_\Theta = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left\{ \|\mathbf{V}(j\omega) - \mathbf{W}(j\omega)\Theta(j\omega)\|_{\Gamma}^2 + \|\Theta(j\omega)\|_{\mathbf{R}}^2 \right\} d\omega \quad (22.6.8)$$

where Γ and $\mathbf{R}$ are nonnegative symmetrical matrices.

We can then apply Parseval's theorem (in subsection §4.10.2) to convert J_Θ into the time domain. The transfer functions in (22.6.6) and (22.6.7) are stable and strictly proper, so this yields

$$J_\Theta = \int_0^\infty \left\{ \|y_1(t) - y_2(t)\|_{\Gamma}^2 + \|u(t)\|_{\mathbf{R}}^2 \right\} dt \quad (22.6.9)$$

where

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \mathbf{B}_2 \end{bmatrix} u(t); \quad \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} \mathbf{B}_1 \\ \mathbf{0} \end{bmatrix} \quad (22.6.10)$$

$$\begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix} = \begin{bmatrix} \mathbf{C}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} \quad (22.6.11)$$

Using (22.6.10) and (22.6.11) in (22.6.9), we obtain

$$J_{\Theta} = \int_0^{\infty} \{x(t)^T \Psi x(t) + u(t)^T R u(t)\} dt \quad (22.6.12)$$

where $x(t) = [x_1(t)^T \ x_2(t)^T]^T$ and

$$\Psi = \begin{bmatrix} C_1^T \\ -C_2^T \end{bmatrix} \Gamma [C_1 \ -C_2] \quad (22.6.13)$$

We recognize (22.6.10)-(22.6.12) as a standard LQR problem, as in section §22.4, where

$$A = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix}; \quad B = \begin{bmatrix} 0 \\ B_2 \end{bmatrix} \quad (22.6.14)$$

Remark 22.1. Note that, to achieve the transformation of the model-matching problem into a LQR problem, the key step is to link $\mathcal{L}^{-1}[\Theta(s)]$ to $u(t)$.

22.6.3 Solution

We are interested in expressing $u(t)$ as a function of $x(t)$ —i.e.,

$$u(t) = -Kx(t) = -[K_1 \ K_2] \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} \quad (22.6.15)$$

such that J_{Θ} in (22.6.12) is minimized. The optimal value of K is as in (22.4.14). We will also assume that the values of A , B , Φ , etc. are such that K corresponds to a stabilizing solution.

The final input $u(t)$ satisfies

$$\dot{x}(t) = Ax(t) + Bu(t) \quad x(0) = [B_1^T \ 0]^T \quad (22.6.16)$$

$$u(t) = -Kx(t) \quad (22.6.17)$$

In transfer-function form, this is

$$U(s) = \Theta(s) = -K(sI - A + BK)^{-1} \begin{bmatrix} B_1 \\ 0 \end{bmatrix} \quad (22.6.18)$$

which, upon our using the special structure of A , B , and K , yields

$$\Theta(s) = [-I + K_2(sI - A_2 + B_2 K_2)^{-1} B_2] K_1 (sI - A_1)^{-1} B_1 \quad (22.6.19)$$

22.7 Discrete-Time Optimal Regulators

The theory for optimal quadratic regulators for continuous-time systems can be extended in a straightforward way to provide similar tools for discrete-time systems. We will briefly summarize the main results, without providing proofs—they parallel those for the continuous-time case presented above.

Consider a discrete-time system having the following state space description:

$$x[k+1] = \mathbf{A}_q x[k] + \mathbf{B}_q u[k] \quad (22.7.1)$$

$$y[k] = \mathbf{C}_q x[k] \quad (22.7.2)$$

where $x \in \mathbb{R}^n$, $u \in \mathbb{R}^m$, and $y \in \mathbb{R}^p$.

For simplicity of notation, we will drop the subscript q from the matrices $\mathbf{A}_q$, $\mathbf{B}_q$, and $\mathbf{C}_q$ in the remainder of this section.

Consider now the cost function

$$J_u(x[k_o], k_o) = \sum_{k_o}^{k_f} \left(x[k]^T \Psi x[k] + u[k]^T \Phi u[k] \right) + x[k_f]^T \Psi_f x[k_f] \quad (22.7.3)$$

The optimal quadratic regulator is given by

$$u^o[k] = -\mathbf{K}_u[k] x[k] \quad (22.7.4)$$

where $\mathbf{K}_u[k]$ is a time-varying gain, given by

$$\mathbf{K}_u[k] = (\Phi + \mathbf{B}^T \mathbf{P}[k] \mathbf{B})^{-1} \mathbf{B}^T \mathbf{P}[k] \mathbf{A} \quad (22.7.5)$$

where $\mathbf{P}[k]$ satisfies the following Discrete Time Dynamic Riccati Equation (DTDRE).

$$\mathbf{P}[k] = \mathbf{A}^T \left(\mathbf{P}[k+1] - \mathbf{P}[k+1] \mathbf{B} (\Phi + \mathbf{B}^T \mathbf{P}[k+1] \mathbf{B})^{-1} \mathbf{B}^T \mathbf{P}[k+1] \right) \mathbf{A} + \Psi \quad (22.7.6)$$

Note that this equation must also be solved backwards, subject to the boundary condition

$$P[k_f] = \Psi_f \quad (22.7.7)$$

The steady-state ($k_f \rightarrow \infty$) version of the control law (22.7.4) is given by

$$u^o[k] = -K_\infty x[k] \quad \text{where} \quad K_\infty = (\Phi + B^T P_\infty B)^{-1} B^T P_\infty A \quad (22.7.8)$$

where K_∞ and P_∞ satisfy the associated Discrete Time Algebraic Riccati Equation (DTARE):

$$A^T \left(P_\infty - P_\infty B (\Phi + B^T P_\infty B)^{-1} B^T P_\infty \right) A + \Psi - P_\infty = 0 \quad (22.7.9)$$

with the property that $A - BK_\infty$ has all its eigenvalues inside the stability boundary, provided that (A, B) is stabilizable and $(A, \Psi^{\frac{1}{2}})$ has no unobservable modes on the unit circle.

22.8 Connections to Pole Assignment

Note that, under reasonable conditions, the steady-state LQR ensures closed-loop stability. However, the connection to the precise closed-loop dynamics is rather indirect; it depends on the choice of Ψ and Φ . Thus, in practice, one usually needs to perform some trial-and-error procedure to obtain satisfactory closed-loop dynamics.

In some circumstances, it is possible to specify a region in which the closed-loop poles should reside and to enforce this in the solution. A simple example of this is when we require that the closed-loop poles have real part to the left of $s = -\alpha$, for $\alpha \in \mathbb{R}^+$. This can be achieved by first shifting the axis by the transformation

$$v = s + \alpha \quad (22.8.1)$$

Then $\Re\{s\} = -\alpha \implies \Re\{v\} = 0$.

To illustrate the application of this transformation, the Laplace form of (22.2.1) is

$$sX(s) = A_o X(s) + B_o U(s) \quad (22.8.2)$$

whilst in the v form it is

$$vX(v) = (\mathbf{A}_o + \alpha\mathbf{I})X(v) + \mathbf{B}_oU(v) = \mathbf{A}_o'X(v) + \mathbf{B}_oU(v) \quad (22.8.3)$$

Hence, if the closed-loop poles are in the left plane of $\boxed{v}$, they will be to the left of α in $\boxed{s}$. Note, however, that this procedure does not enforce any minimal damping on the poles. This is a nice example of a technique that ensures a particular constructed property (real part of closed-loop poles to the left of a given region) but says nothing about the consequential property of damping.

A slightly more interesting demand is to require that the closed-loop poles lie inside a circle with radius ρ and with center at $(-\alpha, 0)$, with $\alpha > \rho \geq 0$ —i.e., the circle is entirely within the LHP.

This can be achieved by using a two-step procedure:

- (i) We first transform the Laplace variable s to a new variable, ζ , defined as follows:

$$\zeta = \frac{s + \alpha}{\rho} \quad (22.8.4)$$

This takes the original circle in $\boxed{s}$ to a unit circle in $\boxed{\zeta}$. The corresponding transformed state space model has the form

$$\zeta X(\zeta) = \frac{1}{\rho}(\alpha\mathbf{I} + \mathbf{A}_o)X(\zeta) + \frac{1}{\rho}\mathbf{B}_oU(\zeta) \quad (22.8.5)$$

- (ii) One then treats (22.8.5) as the state space description of a *discrete-time system*. So, solving the corresponding discrete optimal control problem leads to a feedback gain $\mathbf{K}$ such that $\frac{1}{\rho}(\alpha\mathbf{I} + \mathbf{A}_o - \mathbf{B}_o\mathbf{K})$ has all its eigenvalues inside the unit disk. This in turn implies that, when the same control law is applied in continuous time, then the closed-loop poles reside in the original circle in $\boxed{s}$.

Example 22.3. Consider a 2×2 multivariable system having the state space model

$$\mathbf{A}_o = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 0 \\ 3 & -2 & 2 \end{bmatrix}; \quad \mathbf{B}_o = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 2 & -1 \end{bmatrix}; \quad \mathbf{C}_o = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}; \quad \mathbf{D}_o = \mathbf{0} \quad (22.8.6)$$

Find a state-feedback gain matrix $\mathbf{K}$ such that the closed-loop poles are all located in the disk with center at $(-\alpha; 0)$ and with radius ρ , where $\alpha = 6$ and $\rho = 2$.

Solution

We use the approach proposed above: we transform the complex variable s according to (22.8.4) and then solve a discrete-time optimal-regulator problem.

We first need the state space representation in the transformed space. This is evaluated by applying equation (22.8.5), which leads to

$$\mathbf{A}_\zeta = \frac{1}{\rho}(\alpha\mathbf{I} + \mathbf{A}_o) \quad \text{and} \quad \mathbf{B}_\zeta = \frac{1}{\rho}\mathbf{B}_o \quad (22.8.7)$$

The MATLAB command *dlqr*, with weighting matrices $\Psi = \mathbf{I}_3$ and $\Phi = \mathbf{I}_2$, is then used to obtain the optimal gain $\mathbf{K}_\zeta$, which is

$$\mathbf{K}_\zeta = \begin{bmatrix} 7.00 & -4.58 & 7.73 \\ 3.18 & 7.02 & -4.10 \end{bmatrix} \quad (22.8.8)$$

When this optimal gain is used in the original continuous-time system, the closed-loop poles, computed from $\det(s\mathbf{I} - \mathbf{A}_o + \mathbf{B}_o\mathbf{K}_\zeta) = 0$, are located at -5.13 , -5.45 , and -5.59 . All these poles lie in the prescribed region, as expected.

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Note that the above ideas can be extended to other cases in which the desired region can be transformed into the stability region for either the continuous- or discrete-time case by means of a suitable rational transformation. The reader is encouraged to explore other possibilities.

22.9 Observer Design

Next, we turn to the problem of state estimation. Here, we seek a matrix $\mathbf{J} \in \mathbb{R}^{n \times p}$ such that $\mathbf{A} - \mathbf{J}\mathbf{C}$ has its eigenvalues inside the stability region. Again, it is convenient to use quadratic optimization.

As a first step, we note that an observer can be designed for the pair $(\mathbf{C}, \mathbf{A})$ by simply considering an equivalent (called dual) control problem for the pair $(\mathbf{A}, \mathbf{B})$. To illustrate how this is done, consider the *dual* system with

$$\mathbf{A}' = \mathbf{A}^T \quad \mathbf{B}' = \mathbf{C}^T \quad (22.9.1)$$

Then, using any method for state-feedback design, we can find a matrix $\mathbf{K}' \in \mathbb{R}^{p \times n}$ such that $\mathbf{A}' - \mathbf{B}'\mathbf{K}'$ has its eigenvalues inside the stability region. Hence, if we choose $\mathbf{J} = (\mathbf{K}')^T$, then we have ensured that $\mathbf{A} - \mathbf{J}\mathbf{C}$ has its eigenvalues inside the stability region. Thus, we have completed the observer design.

The procedure leads to a stable state estimation of the form

$$\dot{\hat{x}}(t) = \mathbf{A}_o\hat{x}(t) + \mathbf{B}_ou(t) + \mathbf{J}(y(t) - \mathbf{C}\hat{x}(t)) \quad (22.9.2)$$

Of course, using the tricks outlined above for state-variable feedback, one can also use transformation techniques to ensure that the poles of the observer end up in any region that can be related to either the continuous- or the discrete-time case by a rational transformation.

We will show how the above procedure can be formalized by using Optimal Filtering theory. The resulting optimal filter is called a Kalman filter, to honor Kalman's seminal contribution to this problem.

22.10 Linear Optimal Filters

We will present two alternative derivations of the optimal filter—one based on stochastic modeling of the noise (subsection §22.10.1), and one based on deterministic assumptions (subsection §22.10.2). Readers can choose either derivations or both, depending upon the particular point of view they wish to adopt.

22.10.1 Derivation Based on a Stochastic Noise Model

In this section, we show how optimal-filter design can be set up as a quadratic optimization problem. This shows that the filter is *optimal* under certain assumptions regarding the signal-generating mechanism. In practice, this property is probably less important than the fact that the resultant filter has the right kind of *tuning knobs* so that it can be flexibly applied to a large range of problems of practical interest. The reader is encouraged to understand this core idea of the Kalman filter; it is arguably one of the most valuable tools in the control-system designer's *tool box*.

Consider a linear stochastic system of the form

$$dx(t) = Ax(t)dt + dw(t) \quad (22.10.1)$$

$$dy(t) = Cx(t)dt + dv(t) \quad (22.10.2)$$

where $dv(t)$ and $dw(t)$ are known as *orthogonal increment processes*. As a formal treatment of stochastic differential equations is beyond the scope of this book, it suffices here to think of the formal notation $\dot{w}(t)$, $\dot{v}(t)$ as white-noise processes with impulsive correlation:

$$\mathcal{E}\{\dot{w}(t)\dot{w}(\zeta)^T\} = \mathbf{Q}\delta(t - \zeta) \quad (22.10.3)$$

$$\mathcal{E}\{\dot{v}(t)\dot{v}(\zeta)^T\} = \mathbf{R}\delta(t - \zeta) \quad (22.10.4)$$

where $\mathcal{E}\{\circ\}$ denotes mathematical expectation and $\delta(\circ)$ is the Dirac-delta function. We can then *informally* write the model (22.10.1)-(22.10.2) as

$$\frac{dx(t)}{dt} = \mathbf{A}x(t) + \frac{dw(t)}{dt} \quad (22.10.5)$$

$$y'(t) = \frac{dy(t)}{dt} = \mathbf{C}x(t) + \frac{dv(t)}{dt} \quad (22.10.6)$$

We also assume that $\dot{w}(t)$ and $\dot{v}(t)$ are mutually uncorrelated. For readers familiar with the notation of spectral density for random processes, we are simply requiring that the spectral density for $\dot{w}(t)$ and $\dot{v}(t)$ be $\mathbf{Q}$ and $\mathbf{R}$, respectively.

Our objective will be to find a *linear* filter driven by $y'(t)$ that produces a state estimate $\hat{x}(t)$. We will optimize the filter by minimizing the quadratic function

$$J_t = \mathcal{E}\{\tilde{x}(t)\tilde{x}(t)^T\} \quad (22.10.7)$$

where

$$\tilde{x}(t) = \hat{x}(t) - x(t) \quad (22.10.8)$$

is the estimation error.

We will proceed to the solution of this problem in four steps.

Step 1

Consider a time-varying version of the model (22.10.5), given by

$$\frac{dx_z(t)}{dt} = \mathbf{A}_z(t)x_z(t) + \dot{w}_z(t) \quad (22.10.9)$$

$$y'_z(t) = \frac{dy_z(t)}{dt} = \mathbf{C}_z(t)x_z(t) + \dot{v}_z(t) \quad (22.10.10)$$

where $\dot{w}_z(t)$ and $\dot{v}_z(t)$ have zero mean and are uncorrelated, and

$$\mathcal{E}\{\dot{w}_z(t)\dot{w}_z(\zeta)^T\} = \mathbf{Q}_z(t)\delta(t - \zeta) \quad (22.10.11)$$

$$\mathcal{E}\{\dot{v}_z(t)\dot{v}_z(\zeta)^T\} = \mathbf{R}_z(t)\delta(t - \zeta) \quad (22.10.12)$$

For this model, we wish to compute $\bar{\mathbf{P}}(t) = \mathcal{E}\{x_z(t)x_z(t)^T\}$. We assume that $\mathcal{E}\{x_z(0)x_z(0)^T\} = \bar{\mathbf{P}}_0$, with $\dot{w}_z(t)$ uncorrelated with the initial state $x_z(0) = x_{0z}$.

Solution

The solution to (22.10.9) is

$$x_z(t) = \phi_z(t, 0)x_{oz} + \int_0^t \phi_z(t, \tau)\dot{w}_z(\tau)d\tau \quad (22.10.13)$$

where $\phi_z(t_2, t_1) \in \mathbb{R}^{n \times n}$ is the state transition matrix for the system (22.10.9).

Then, squaring (22.10.13) and taking mathematical expectations, we have

$$\bar{\mathbf{P}}(t) = \mathcal{E}\{x_z(t)x_z(t)^T\} = \phi_z(t, 0)\bar{\mathbf{P}}_o\phi_z(t, 0)^T + \int_0^t \phi_z(t, \tau)\mathbf{Q}_z(\tau)\phi_z(t, \tau)^T d\tau \quad (22.10.14)$$

where we have used (22.10.11).

Note that, by applying the Leibnitz rule (3.7.4) to (22.10.14), we obtain

$$\frac{d\bar{\mathbf{P}}(t)}{dt} = \mathbf{A}_z\bar{\mathbf{P}}(t) + \bar{\mathbf{P}}(t)\mathbf{A}_z^T + \mathbf{Q}_z(t) \quad (22.10.15)$$

where we have also used the fact that $\frac{d}{dt}\phi(t, \tau) = \mathbf{A}_z(t)\phi(t, \tau)$.

Step 2

We now return to the original problem: to obtain an estimate, $\hat{x}(t)$, for the state, $x(t)$, in (22.10.5). We assume the following linear form for the filter:

$$\frac{d\hat{x}(t)}{dt} = \mathbf{A}\hat{x}(t) + \mathbf{J}(t)[y'(t) - \mathbf{C}x(t)] \quad (22.10.16)$$

where $\mathbf{J}(t)$ is a time-varying gain yet to be determined. Note that (22.10.16) has the same structure as a standard linear observer, save that it has a time-varying observer gain.

Step 3

Assume that we are also given an initial state estimate $\hat{x}_o$ of statistical property

$$\mathcal{E}\{(x(0) - \hat{x}_o)(x(0) - \hat{x}_o)^T\} = \mathbf{P}_o, \quad (22.10.17)$$

and assume, for the moment, that we are given some gain $\mathbf{J}(\tau)$ for $0 \leq \tau \leq t$. Derive an expression for

$$\begin{aligned} \mathbf{P}(t) &= \mathcal{E}\{(\hat{x}(t) - x(t))(\hat{x}(t) - x(t))^T\} \\ &= \mathcal{E}\{\tilde{x}(t)\tilde{x}(t)^T\} \end{aligned} \quad (22.10.18)$$

Solution

Subtracting (22.10.16) from (22.10.5), we obtain

$$\frac{d\tilde{x}(t)}{dt} = (\mathbf{A} - \mathbf{J}(t)\mathbf{C})\tilde{x}(t) + \mathbf{J}(t)\dot{v}(t) - \dot{w}(t) \quad (22.10.19)$$

We see that (22.10.19) is a time-varying system, and we can therefore immediately apply the solution to Step 1, after making the following connections:

$$x_z(t) \rightarrow \tilde{x}(t); \quad \mathbf{A}_z(t) \rightarrow (\mathbf{A} - \mathbf{J}(t)\mathbf{C}); \quad \dot{w}_z(t) \rightarrow \mathbf{J}(t)\dot{v}(t) - \dot{w}(t) \quad (22.10.20)$$

to conclude

$$\frac{d\mathbf{P}(t)}{dt} = (\mathbf{A} - \mathbf{J}(t)\mathbf{C})\mathbf{P}(t) + \mathbf{P}(t)(\mathbf{A} - \mathbf{J}(t)\mathbf{C})^T + \mathbf{J}(t)\mathbf{R}\mathbf{J}(t)^T + \mathbf{Q} \quad (22.10.21)$$

subject to $\mathbf{P}(0) = \mathbf{P}_0$. Note that we have used the fact that $\mathbf{Q}_z(t) = \mathbf{J}(t)\mathbf{R}\mathbf{J}(t)^T + \mathbf{Q}$.

Step 4

We next choose $\mathbf{J}(t)$, at each time instant, so that $\dot{\mathbf{P}}$ is as small as possible.

Solution. We complete the square on the right-hand side of (22.10.21) by defining $\mathbf{J}(t) = \mathbf{J}^*(t) + \tilde{\mathbf{J}}(t)$ where $\mathbf{J}^*(t) = \mathbf{P}(t)\mathbf{C}^T\mathbf{R}^{-1}$. Substituting into (22.10.21) gives

$$\begin{aligned} \frac{d\mathbf{P}(t)}{dt} &= (\mathbf{A} - \mathbf{J}(t)\mathbf{C} - \tilde{\mathbf{J}}(t)\mathbf{C})\mathbf{P}(t) + \mathbf{P}(t)(\mathbf{A} - \mathbf{J}(t)\mathbf{C} - \tilde{\mathbf{J}}(t)\mathbf{C})^T \\ &\quad + (\mathbf{J}^*(t) + \tilde{\mathbf{J}}(t))\mathbf{R}(\mathbf{J}^*(t) + \tilde{\mathbf{J}}(t))^T + \mathbf{Q} \\ &= (\mathbf{A} - \mathbf{J}(t)\mathbf{C})\mathbf{P}(t) + \mathbf{P}(t)(\mathbf{A} - \mathbf{J}(t)\mathbf{C})^T \\ &\quad + \mathbf{J}^*(t)\mathbf{R}\mathbf{J}^*(t) + \mathbf{Q} + \tilde{\mathbf{J}}(t)\mathbf{R}(\tilde{\mathbf{J}}(t))^T \end{aligned} \quad (22.10.22)$$

We clearly see that $\dot{\mathbf{P}}(t)$ is minimized at every time if we choose $\tilde{\mathbf{J}}(t) = 0$. Thus, $\mathbf{J}^*(t)$ is the optimal-filter gain, because it minimizes $\dot{\mathbf{P}}(t)$ (and hence $\mathbf{P}(t)$) for all t .

□□□

In summary, the optimal filter satisfies

$$\frac{d\hat{x}(t)}{dt} = \mathbf{A}\hat{x}(t) + \mathbf{J}^*(t)[y'(t) - \mathbf{C}\hat{x}(t)] \quad (22.10.23)$$

where the optimal gain $\mathbf{J}^*(t)$ satisfies

$$\mathbf{J}^*(t) = \mathbf{P}(t)\mathbf{C}^T\mathbf{R}^{-1} \quad (22.10.24)$$

and $\mathbf{P}(t)$ is the solution to

$$\begin{aligned} \frac{d\mathbf{P}(t)}{dt} = & (\mathbf{A} - \mathbf{J}^*(t)\mathbf{C})\mathbf{P}(t) + \mathbf{P}(t)(\mathbf{A} - \mathbf{J}^*(t)\mathbf{C})^T \\ & + \mathbf{J}^*(t)\mathbf{R}(\mathbf{J}^*(t))^T + \mathbf{Q} \end{aligned} \quad (22.10.25)$$

subject to $\mathbf{P}(0) = \mathbf{P}_0$.

Equation (22.10.25) can also be simplified to

$$\frac{d\mathbf{P}(t)}{dt} = \mathbf{Q} - \mathbf{P}(t)\mathbf{C}^T\mathbf{R}^{-1}\mathbf{C}\mathbf{P}(t) + \mathbf{P}(t)\mathbf{A}^T + \mathbf{A}\mathbf{P}(t) \quad (22.10.26)$$

The reader will recognize that the solution to the optimal linear filtering problem presented above has a very close connection to the LQR problem presented in section §22.4.

Remark 22.2. *It is important to note, in the above derivation, that it makes no difference whether the system is time varying (i.e., $\mathbf{A}$, $\mathbf{C}$, $\mathbf{Q}$, $\mathbf{R}$, etc. are all functions of time). This is often important in applications.*

Remark 22.3. *When we come to properties, these are usually restricted to the time-invariant case (or closely related cases—e.g., periodic systems). Thus, when discussing the steady-state filter, it is usual to restrict attention to the case in which $\mathbf{A}$, $\mathbf{C}$, $\mathbf{Q}$, $\mathbf{R}$, etc. are not explicit functions of time.*

The properties of the optimal filter therefore follow directly from the optimal LQR solutions, under the correspondences given in Table 22.10.1 below.

In particular, one is frequently interested in the steady-state optimal filter obtained when $\mathbf{A}$, $\mathbf{C}$, $\mathbf{Q}$ and $\mathbf{R}$ are time invariant and the filtering horizon tends to infinity. By duality with the optimal control problem, the steady-state filter takes the form

$$\frac{d\hat{\mathbf{x}}(t)}{dt} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{J}_s^\infty(\mathbf{y}' - \mathbf{C}\hat{\mathbf{x}}) \quad (22.10.27)$$

Regulator	Filter
τ	$t - \tau$
t_f	0
A	$-A^T$
B	$-C^T$
Ψ	Q
Φ	R
Ψ_f	P_o

Table 22.1. Duality between quadratic regulators and filters

where

$$J_s^\infty = P_\infty^s C^T R^{-1} \quad (22.10.28)$$

and P_∞^s is the stabilizing solution of the following CTARE:

$$Q - P_\infty C^T R^{-1} C P_\infty + P_\infty A^T + A P_\infty = 0 \quad (22.10.29)$$

We will not labor the time-variant properties here; they are easily obtained from the corresponding properties of the optimal regulator. To illustrate, we state without proof the following facts that are the duals of those given in subsection §22.5.1 for the optimal regulator:

- (i) Say that the system (C, A) is detectable from $y(t)$ (the reader may judge that this assumption is reasonable—after all, if the system isn't detectable, then there are *unstable states* whose response cannot be *seen* in the output, and it is heuristically reasonable that it would then be impossible to say anything sensible about these states by processing the output signal); and
- (ii) say that the system states are all perturbed by noise, so that it is not possible to estimate them once and for all, and then let the open-loop model predict

all future values. (Technically, this is stated as requiring that $(\mathbf{A}, \mathbf{Q}^{\frac{1}{2}})$ is stabilizable.)

Then, the optimal solution of the filtering Riccati equation (22.10.26) tends to a steady-state limit $\mathbf{P}_{\infty}^s$ as $t \rightarrow \infty$. This limit has two key properties:

- $\mathbf{P}_{\infty}^s$ is the only nonnegative solution of the matrix algebraic Riccati Equation (22.10.29) obtained by setting $\frac{d\mathbf{P}(t)}{dt}$ in (22.10.26).
- When this steady-state value is used to generate a steady-state observer, as in $\{(22.10.27), (22.10.28)\}$, then the observer has the property that $(\mathbf{A} - \mathbf{J}_{\infty}^s \mathbf{C})$ is a stability matrix.

This last property is important in applications. In particular, we recall that the error $\tilde{x}(t)$ between $\hat{x}(t)$ and $x(t)$ satisfies (22.10.19). The steady-state version of this equation is stable when $(\mathbf{A} - \mathbf{J}_{\infty}^s \mathbf{C})$ is a stability matrix.

22.10.2 Steady-State Filter as a Deterministic Model-Matching Problem

We next give an alternative, purely *deterministic* view of the *optimal* filter as a model-matching problem. Consider a state space model, as in $\{(22.10.1), (22.10.2)\}$ without noise:

$$\frac{dx(t)}{dt} = \mathbf{A}x(t); \quad x(0) = x_o \quad (22.10.30)$$

$$y_o(t) = \mathbf{C}x(t) \quad (22.10.31)$$

Say that we are interested in estimating a scalar combination of the state, which we denote by

$$z_o(t) = \gamma^T x(t) \quad (22.10.32)$$

or, in transfer-function form

$$Z_o(s) = \gamma^T (s\mathbf{I} - \mathbf{A})^{-1} x_o \quad (22.10.33)$$

Our filter is required to be a linear time-invariant system driven by y_o . This we express in transfer-function form by requiring that

$$\hat{Z}_o(s) = \mathbf{F}(s)Y_o(s) \quad (22.10.34)$$

where $\mathbf{F}(s)$ is the $1 \times m$ row vector corresponding to the filter transfer function and

$$Y_o(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}x_o \quad (22.10.35)$$

Our remaining task is to find the optimal value for the transfer function $\mathbf{F}(s)$. Toward this goal, we want to consider all possible initial states x_o . We know that all such states can be formed as a linear combination of the basis vectors $e_1, \dots, e_n$, where $e_i = [0, \dots, 0, 1, 0, \dots, 0]^T$. Thus we vectorize (22.10.33), (22.10.34), (22.10.35) by changing x_o into each of the above vectors in turn—i.e., we introduce the row vectors

$$\mathbf{Z} = [z_1, \dots, z_n] \quad (22.10.36)$$

$$\hat{\mathbf{Z}} = [\hat{z}_1, \dots, \hat{z}_n] \quad (22.10.37)$$

and the $m \times n$ matrix

$$\mathbf{Y} = [y_1, \dots, y_n] \quad (22.10.38)$$

where

$$Z_i(s) = \gamma^T (s\mathbf{I} - \mathbf{A})^{-1} e_i \quad (22.10.39)$$

$$Y_i(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1} e_i \quad (22.10.40)$$

$$\hat{Z}_i(s) = \mathbf{F}(s)\mathbf{Y}_i(s) \quad (22.10.41)$$

We would like the row vector $\hat{\mathbf{Z}}(s)$ to be *near* $\mathbf{Z}(s)$ in some sense. However, we wish to place some weighting on the size of the filter to avoid large values of $\mathbf{F}$ that would magnify high-frequency components in measurement disturbances.

A criterion that trades off the closeness of $\mathbf{Z}$ to $\hat{\mathbf{Z}}$ with the size of $\mathbf{F}$ is the following quadratic criterion in the frequency domain:

$$J = \int_{-\infty}^{\infty} \left(\|\mathbf{Z}(j\omega)^T - \hat{\mathbf{Z}}(j\omega)^T\|_{\mathbf{Q}}^2 + \|\mathbf{F}(j\omega)\|_{\mathbf{R}}^2 \right) d\omega \quad (22.10.42)$$

where we have somewhat arbitrarily introduced weightings $\mathbf{Q}$ and $\mathbf{R}$ to allow us to shift the balance between bringing $\hat{\mathbf{Z}}$ closer to $\mathbf{Z}$ versus having a large filter gain. Using (22.10.33), (22.10.34), (22.10.35), the criterion becomes

$$J = \int_{-\infty}^{\infty} \left(\|\mathbf{Z}^T - (\mathbf{F}\mathbf{Y})^T\|_{\mathbf{Q}}^2 + \|\mathbf{F}^T\|_{\mathbf{R}}^2 \right) d\omega \quad (22.10.43)$$

It is interesting to transpose quantities in the above expression. This preserves the norm but shifts $\mathbf{F}$ to an intuitively interesting position. Doing this yields

$$J = \int_{-\infty}^{\infty} \left(\|\mathbf{Z}^T - \mathbf{Y}^T \mathbf{F}^T\|_{\mathbf{Q}}^2 + \|\mathbf{F}^T\|_{\mathbf{R}}^2 \right) d\omega \quad (22.10.44)$$

We can now think of $\mathbf{F}^T$ as a control vector. In this form, we recognize (22.10.44) as being simply a standard model-matching problem, exactly as was discussed in section §22.6, with the correspondences

$$\mathbf{Y}^T = (s\mathbf{I} - \mathbf{A}^T)^{-1}\mathbf{C}^T \equiv \mathbf{W} = \mathbf{C}_2(s\mathbf{I} - \mathbf{A}_2)^{-1}\mathbf{B}_2 \quad (22.10.45)$$

$$\mathbf{Z}^T = (s\mathbf{I} - \mathbf{A}^T)^{-1}\gamma \equiv \mathbf{V} = \mathbf{C}_1(s\mathbf{I} - \mathbf{A}_1)^{-1}\mathbf{B}_1 \quad (22.10.46)$$

$$(22.10.47)$$

where $\mathbf{C}_2 = \mathbf{C}_1 = \mathbf{I}$, $\mathbf{A}_1 = \mathbf{A}_2 = \mathbf{A}^T$, $\mathbf{B}_2 = \mathbf{C}^T$, and $\mathbf{B}_1 = \gamma$.

From section §22.6, the optimal solution is

$$\mathbf{F}(s)^T = -[\mathbf{I} - \mathbf{K}_2(s\mathbf{I} - \mathbf{A}^T + \mathbf{C}^T\mathbf{K}_2)^{-1}\mathbf{C}^T][\mathbf{K}_1(s\mathbf{I} - \mathbf{A}^T)^{-1}\gamma] \quad (22.10.48)$$

or

$$\mathbf{F}(s) = \gamma^T[s\mathbf{I} - \mathbf{A}]^{-1}\mathbf{K}_1^T[-\mathbf{I} + \mathbf{C}(s\mathbf{I} - \mathbf{A} + \mathbf{K}_2^T\mathbf{C})^{-1}\mathbf{K}_2^T] \quad (22.10.49)$$

Now, because of the special structure of the problem, the Riccati equation and optimal gains have a special form. Specifically, we have that the Riccati equation is

$$0 = \tilde{\Psi} - \tilde{\mathbf{P}}\tilde{\mathbf{B}}\tilde{\Phi}^{-1}\tilde{\mathbf{B}}^T\tilde{\mathbf{P}} + \tilde{\mathbf{P}}\tilde{\mathbf{A}} + \tilde{\mathbf{A}}^T\tilde{\mathbf{P}} \quad (22.10.50)$$

which, upon using the special structure of $\tilde{\mathbf{A}}$, $\tilde{\mathbf{B}}$, $\tilde{\Psi}$, and $\tilde{\Phi}$, gives

$$0 = \begin{bmatrix} \mathbf{Q} & -\mathbf{Q} \\ -\mathbf{Q} & \mathbf{Q} \end{bmatrix} - \begin{bmatrix} \mathbf{P}_3\mathbf{C}^T\mathbf{R}^{-1}\mathbf{C}\mathbf{P}_3 & \mathbf{P}_3\mathbf{C}\mathbf{R}^{-1}\mathbf{C}\mathbf{P}_2 \\ \mathbf{P}_2\mathbf{C}^T\mathbf{R}^{-1}\mathbf{C}\mathbf{P}_3^T & \mathbf{P}_2\mathbf{C}^T\mathbf{R}^{-1}\mathbf{C}\mathbf{P}_2 \end{bmatrix} \quad (22.10.51)$$

$$+ \begin{bmatrix} \mathbf{P}_1\mathbf{A}^T & \mathbf{P}_3\mathbf{A}^T \\ \mathbf{P}_3^T\mathbf{A}^T & \mathbf{P}_2\mathbf{A}^T \end{bmatrix} + \begin{bmatrix} \mathbf{A}\mathbf{P}_1 & \mathbf{A}\mathbf{P}_3 \\ \mathbf{A}\mathbf{P}_3^T & \mathbf{A}\mathbf{P}_2 \end{bmatrix}$$

where we have used

$$\tilde{\mathbf{A}} = \begin{bmatrix} \mathbf{A}_1^T & 0 \\ 0 & \mathbf{A}_2^T \end{bmatrix}; \quad \tilde{\mathbf{B}} = \begin{bmatrix} 0 \\ \mathbf{C}^T \end{bmatrix}; \quad \mathbf{P} = \begin{bmatrix} \mathbf{P}_1 & \mathbf{P}_3 \\ \mathbf{P}_3^T & \mathbf{P}_2 \end{bmatrix}; \quad \tilde{\Phi} = \mathbf{R} \quad (22.10.52)$$

We note that, in our problem, $\mathbf{A}_1 = \mathbf{A}_2 = \mathbf{A}^T$ and $\mathbf{B}_2 = \mathbf{C}^T$ and that $\bar{\mathbf{P}} = \mathbf{P}_2$ satisfies the following separate equation:

$$0 = \mathbf{Q} - \bar{\mathbf{P}}\mathbf{C}^T\mathbf{R}^{-1}\mathbf{C}\bar{\mathbf{P}} + \bar{\mathbf{P}}\mathbf{A}^T + \mathbf{A}\bar{\mathbf{P}} \quad (22.10.53)$$

Also, we have that P_3 satisfies

$$0 = -Q - \bar{P}C^T R^{-1} C P_3^T + P_3^T A^T + A P_3^T \quad (22.10.54)$$

Adding (22.10.53) and (22.10.54) gives

$$-\bar{P}C^T R^{-1} C (P_3^T + \bar{P}) + (P_3^T + \bar{P})A^T + A(P_3^T + \bar{P}) = 0 \quad (22.10.55)$$

which leads to $P_3^T = -\bar{P} = P_3$

Finally,

$$\begin{aligned} [K_1 \quad K_2] &= \Phi^{-1} [0 \quad B_2^T] \begin{bmatrix} P_1 & P_3 \\ P_3 & P_2 \end{bmatrix} \\ &= \Phi^{-1} [B_2^T P_3 \quad B_2^T P_2] \\ &= [-R^{-1} C \bar{P} \quad R^{-1} C \bar{P}] \end{aligned} \quad (22.10.56)$$

Setting $J^T = -K_1 = K_2$, then,

$$J = \bar{P}C^T R^{-1} \quad (22.10.57)$$

Finally, substituting (22.10.57) into (22.10.49), we have

$$\begin{aligned} F(s) &= \gamma^T [sI - A]^{-1} J [I - C(sI - A + JC)^{-1} J] \\ &= \gamma^T [sI - A]^{-1} [I - JC(sI - A + JC)^{-1}] J \\ &= \gamma^T (sI - A)^{-1} [sI - A + JC - JC] (sI - A + JC)^{-1} J \\ &= \gamma^T [sI - A + JC]^{-1} J \end{aligned} \quad (22.10.58)$$

We notice that γ^T appears as a factor outside of the filter. Thus, we have actually found a result that can be translated to one which is independent of the choice of γ . In particular, we note from (22.10.32) and (22.10.34), that the optimal filter for estimating x is simply $(sI - A + JC)^{-1} J$. This transfer function can be realized in state space form as

$$\dot{\hat{x}}(t) = (A - JC)\hat{x}(t) + Jy(t) \quad (22.10.59)$$

which we recognize as being identical to the steady-state filter given in (22.10.27).

Remark 22.4. *The above derivation gives a purely deterministic view of the optimal filtering and allows us to trade off the size of the error versus the size of the filter.*

Other connections with deterministic optimal control theory are explored in §D.5 of the Appendix.

□□□

22.10.3 Discrete-Time Optimal Quadratic Filter

As we did in section §22.7 for the control problem, we can readily develop discrete forms for the optimal filter. As in section §22.7, we will simply quote the result; the theory parallels the continuous case.

Consider a discrete-time system having the following state space description:

$$x[k+1] = \mathbf{A}x[k] + \mathbf{B}u[k] + w[k] \quad (22.10.60)$$

$$y[k] = \mathbf{C}[k] + v[k] \quad (22.10.61)$$

where $w[k] \in \mathbb{R}^n$ and $v[k] \in \mathbb{R}^p$ are uncorrelated stationary stochastic processes, with covariances given by

$$\mathcal{E}\{w[k]w^T[\zeta]\} = \mathbf{Q}\delta_K[k - \zeta] \quad (22.10.62)$$

$$\mathcal{E}\{v[k]v^T[\zeta]\} = \mathbf{R}\delta_K[k - \zeta] \quad (22.10.63)$$

where $\mathbf{Q} \in \mathbb{R}^{n \times n}$ is a symmetric nonnegative definite matrix and $\mathbf{R} \in \mathbb{R}^{p \times p}$ is a symmetric positive definite matrix.

Consider now the following observer to estimate the system state:

$$\hat{x}[k+1] = \mathbf{A}\hat{x}[k] + \mathbf{B}u[k] + \mathbf{J}_o[k](y[k] - \mathbf{C}\hat{x}[k]) \quad (22.10.64)$$

Furthermore, assume that the initial state $x[0]$ satisfies

$$\mathcal{E}\{(x[0] - \hat{x}[0])(x[0] - \hat{x}[0])^T\} = \mathbf{P}_o \quad (22.10.65)$$

Then the optimal choice (in a quadratic sense) for $\mathbf{J}_o[k]$ is given by

$$\mathbf{J}_o[k] = \mathbf{A}\mathbf{P}[k]\mathbf{C}^T(\mathbf{R} + \mathbf{C}\mathbf{P}[k]\mathbf{C}^T)^{-1} \quad (22.10.66)$$

where $P[k]$ satisfies the following discrete-time dynamic Riccati equation (DTDRE).

$$P[k+1] = A(P[k] - P[k]C^T(R + CP[k]C^T)^{-1}CP[k])A^T + Q \quad (22.10.67)$$

which can be solved forward in time, subject to

$$P[0] = P_0 \quad (22.10.68)$$

Remark 22.5. Note again that A, C, Q, R , etc. can all be time varying for the general filter. However, when we come to discuss the steady-state filter, it is usually required that the problem be time invariant (or, at least, transferable to this form, as is the case for periodic systems).

The steady-state ($k \rightarrow \infty$) filter gain satisfies the DTARE given by

$$A[P_\infty - P_\infty C^T(R + CP_\infty C^T)^{-1}CP_\infty]A^T + Q = P_\infty \quad (22.10.69)$$

The above results show that the duality between optimal regulators and optimal filters is also present in discrete-time systems.

22.10.4 Stochastic Noise Models

In the above development, we have simply represented the noise as a white-noise sequence ($\{\omega(k)\}$) and a white measurement-noise sequence ($\{v(k)\}$). Actually, this is much more general than it may seem at first sight. For example, it can include *colored* noise having an arbitrary rational noise spectrum. The essential idea is to model this noise as the output of a linear system (i.e., a filter) driven by white noise. Thus, say that a system is described by

$$x(k+1) = Ax(k) + Bu(k) + \omega_c(k) \quad (22.10.70)$$

$$y(k) = Cx(k) + v(k) \quad (22.10.71)$$

where $\{\omega_c(k)\}$ represents *colored noise*—noise that is white noise passed through a filter. Then we can add the additional noise model $\{(22.10.70), (22.10.71)\}$ to the description. For example, let the noise filter be

$$x'(k+1) = A'x(k) + \omega(k) \quad (22.10.72)$$

$$\omega_c(k) = C'x'(k) \quad (22.10.73)$$

where $\{\omega(k)\}$ is a white-noise sequence.

Combining (22.10.70) to (22.10.73) gives a composite system *driven by white noise*, of the form

$$\bar{x}(k+1) = \bar{A}\bar{x}(k) + \bar{B}u(k) + \bar{\omega}(k) \quad (22.10.74)$$

$$y(k) = \bar{C}\bar{x}(k) + v(k) \quad (22.10.75)$$

where

$$\bar{x}(k) = [x(k)^T, x'(k)^T]^T \quad (22.10.76)$$

$$\bar{\omega}(k) = [0, \omega(k)^T] \quad (22.10.77)$$

$$\bar{A} = \begin{bmatrix} A & C' \\ 0 & A' \end{bmatrix}; \quad \bar{B} = \begin{bmatrix} B \\ 0 \end{bmatrix}; \quad \bar{C} = [C \quad 0] \quad (22.10.78)$$

22.10.5 Optimal Prediction

In the case of gaussian noise, the Kalman filter actually produces the mean of the conditional distribution for $x(k)$ given the past data: $\{y(\ell), u(\ell); \ell = 0, \dots, k-2, k-1\}$. This can be written as

$$\begin{aligned} \hat{x}(k) &= E\{x(k)/y(\ell), u(\ell); \ell = 0, \dots, k-1\} \\ &\triangleq \hat{x}(k|k-1) \end{aligned} \quad (22.10.79)$$

The unpredictability of white noise causes the optimal prediction of future states to be obtained by simply iterating the model with the future white noise set equal to its expected value, which is zero. Thus, consider the composite model $\{(22.10.74), (22.10.75)\}$, then for a known input $\{u(k)\}$, the predicted future states satisfy

$$\tilde{x}(j+1) = \bar{A}\tilde{x}(j) + \bar{B}u(j); \quad j = k, k+1, k+2, \dots \quad (22.10.80)$$

with initial condition $\tilde{x}(k) = \hat{x}(k|k-1)$. Note that $\tilde{x}(j)$ can also be written as

$$\begin{aligned} \tilde{x}(j) &= E\{x(j)|y(\ell), u(\ell); \ell = 0, \dots, k-1\}; \quad j \geq k \\ &\triangleq \hat{x}(j|k-1); \quad j \geq k \end{aligned} \quad (22.10.81)$$

22.10.6 State Estimation with Continuous Models and Sampled Observations

A problem that is frequently met in practice arises from an underlying continuous-time system whose data is collected at discrete sample points. Indeed, the sample points can, in some applications, be nonuniformly spaced. This still falls under the general discrete time-varying filtering problem.

What we need to do is to propagate the continuous system between samples and then update at the sample points by using an appropriate discrete model. To illustrate, say that the system is described (formally) by (22.10.5) and (22.10.6):

$$\dot{x} = Ax + \dot{w} \quad (22.10.82)$$

$$y' = Cx + \dot{v} \quad (22.10.83)$$

where $\dot{w}$, $\dot{v}$ have spectral density Q and R respectively.

Before we can sensibly sample y' , we need first to pass it through an appropriate anti-aliasing filter. One such filter is the so-called *integrate and dump* filter. In this case, the k^{th} sample is given by

$$\bar{y}[k] = \frac{1}{t_k - t_{k-1}} \int_{t_{k-1}}^{t_k} y'(t) dt \quad (22.10.84)$$

where t_k denotes the time of the k^{th} sample.

The discrete observation $\bar{y}[k]$ can be related to a discrete state space model of the form

$$\bar{x}[k+1] = A_q[k] \bar{x}[k] + \bar{w}[k] \quad (22.10.85)$$

$$\bar{y}[k] = C_q[k] \bar{x}[k] + \bar{v}[k] \quad (22.10.86)$$

By integrating the model $\{(22.10.82), (22.10.83)\}$, it can be seen that

$$A_q[k] = e^{A(t_{k+1}-t_k)} \simeq I + A(t_{k+1} - t_k) \quad (22.10.87)$$

$$C_q[k] = \frac{1}{(t_{k+1} - t_k)} \int_{t_k}^{t_{k+1}} C e^{A\tau} d\tau \simeq C \quad (22.10.88)$$

$$E\{\bar{w}[k] \bar{w}[k]^T\} = \bar{Q} \simeq Q(t_{k+1} - t_k) \quad (22.10.89)$$

$$E\{\bar{v}[k] \bar{v}[k]^T\} = \bar{R} \simeq \frac{R}{(t_{k+1} - t_k)} \quad (22.10.90)$$

The discrete model can then be used to obtain the state estimates *at the sample points*. The continuous state can then be interpolated between samples by using the open-loop model

$$\dot{\hat{x}}(t) = A\hat{x}(t); \quad \text{for} \quad t \in [t_k, t_{k+1}] \quad (22.10.91)$$

with initial condition

$$\hat{x}(t_k) = \hat{\hat{x}}[k] \quad (22.10.92)$$

22.11 State-Estimate Feedback

Finally, we can combine the state estimation provided by (22.9.2) with the state-variable feedback determined earlier to yield the following state-estimate feedback-control law:

$$u(t) = -K\hat{x}(t) + \bar{r}(t) \quad (22.11.1)$$

Note that Theorem 18.1 also applies here; the closed-loop poles resulting from the use of (22.11.1) are the union of the eigenvalues that result from the use of the state feedback together with the eigenvalues associated with the observer. Note, in particular, that the proof of Theorem 18.1 does not depend on the scalar nature of the problem.

22.12 Transfer-Function Interpretation

As in section §18.5, the state-estimation feedback law described above can be given a polynomial interpretation.

In particular, Lemma 18.5 on page 539 still holds for the steady-state observer; i.e., $\hat{X}(s)$ can be expressed as

$$\hat{X}(s) = (sI - A_o + JC_o)^{-1}(B_o U(s) + JY(s)) = T_1(s)U(s) + T_2(s)Y(s) \quad (22.12.1)$$

where $T_1(s)$ and $T_2(s)$ are two stable transfer-function matrices given by

$$T_1(s) \triangleq (sI - A_o + JC_o)^{-1}B_o \quad (22.12.2)$$

$$T_2(s) \triangleq (sI - A_o + JC_o)^{-1}J \quad (22.12.3)$$

Thus, the feedback control law based on the state estimate is given by

$$\begin{aligned} U(s) &= -K\hat{X}(s) - \bar{R}(s) \\ &= -KT_1(s)U(s) - KT_2(s)Y(s) + \bar{R}(s) \end{aligned} \quad (22.12.4)$$

This can be written in an equivalent form as

$$\bar{C}_D(s)U(s) = -\bar{C}_N(s)Y(s) + \bar{R}(s) \quad (22.12.5)$$

where $\bar{C}_D(s)$ and $\bar{C}_N(s)$ are stable matrices given by

$$\bar{C}_D(s) = I + K[sI - A_o + JC_o]^{-1}B_o = \frac{\bar{N}_{CD}(s)}{E(s)} \quad (22.12.6)$$

$$\bar{C}_N(s) = K[sI - A_o + JC_o]^{-1}J = \frac{\bar{N}_{CN}(s)}{E(s)} \quad (22.12.7)$$

where $\bar{N}_{CD}(s)$ and $\bar{N}_{CN}(s)$ are polynomial matrices, and where $E(s)$ is the observer polynomial given by

$$E(s) = \det(sI - A_o + JC_o) \quad (22.12.8)$$

From (22.12.5), we see that $C(s) = [\bar{C}_D(s)]^{-1}\bar{C}_N(s)$ is a LMFD for the controller. Furthermore, this controller satisfies the following lemma.

Lemma 22.3. *Consider the controller $C(s) = [\bar{C}_D(s)]^{-1}\bar{C}_N(s)$, with $\bar{C}_D(s)$ and $\bar{C}_N(s)$ defined in (22.12.6) and (22.12.7), respectively. Then a state space realization for this controller is given by the 4-tuple $(A_o - JC_o - B_oK, J, K, 0)$:*

$$C(s) = K[sI - A_o + JC_o + B_oK]^{-1}J \quad (22.12.9)$$

Proof

We first apply the matrix inversion lemma to obtain

$$[\bar{C}_D(s)]^{-1} = I - K[sI - A_o + JC_o + B_oK]^{-1}B_o \quad (22.12.10)$$

We next form the product $[\bar{C}_D(s)]^{-1}\bar{C}_N(s)$ and use (22.12.10) and (22.12.7), from which, after some straightforward matrix manipulations, the result follows. $\square\square\square$

The state space implementation of the controller allows the direct use of the results of the observer and state-feedback synthesis stages. This is useful not only in simulation but also when implementing the controller in practice.

It is also possible to obtain a RMFD for the controller. In particular, from Lemma 20.1 on page 620, it follows that

$$[\bar{C}_D(s)]^{-1}\bar{C}_N(s) = C_N(s)[C_D(s)]^{-1} \quad (22.12.11)$$

where

$$C_D(s) = I - C_o(sI - A_o + B_oK)^{-1}J = \frac{N_{CD}(s)}{F(s)} \quad (22.12.12)$$

$$C_N(s) = K(sI - A_o + B_oK)^{-1}J = \frac{N_{CN}(s)}{F(s)} \quad (22.12.13)$$

where $N_{CD}(s)$ and $N_{CN}(s)$ are polynomial matrices, and where $F(s)$ is the state-feedback polynomial given by

$$F(s) = \det(sI - A_o + B_oK) \quad (22.12.14)$$

Similarly, by using (20.3.24) and (20.3.25) and Lemma 20.1 on page 620, the plant transfer function can be written as

$$\begin{aligned} G_o(s) &= C_o[sI - A_o]^{-1}B_o \\ &= [\bar{G}_{oD}(s)]^{-1}\bar{G}_{oN}(s) = G_{oN}(s)[G_{oD}(s)]^{-1} \end{aligned} \quad (22.12.15)$$

where

$$\bar{G}_{oN}(s) = C_o(sI - A_o + JC_o)^{-1}B_o \quad (22.12.16)$$

$$\bar{G}_{oD}(s) = I - C_o(sI - A_o + JC_o)^{-1}J \quad (22.12.17)$$

$$G_{oN}(s) = C_o(sI - A_o + B_oK)^{-1}B_o \quad (22.12.18)$$

$$G_{oD}(s) = I - K(sI - A_o + B_oK)^{-1}B_o \quad (22.12.19)$$

We then have the following result.

Lemma 22.4. *The sensitivity and complementary sensitivity resulting from the control law 22.12.5 are, respectively,*

$$S_o(s) = C_D(s)\bar{G}_{oD}(s) = I - G_{oN}(s)\bar{C}_N(s) \quad (22.12.20)$$

$$T_o(s) = G_{oN}(s)\bar{C}_N(s) = I - C_D(s)\bar{G}_{oD}(s) \quad (22.12.21)$$

Proof

From Lemma 20.1 on page 620, we have that

$$\bar{G}_{oD}(s)C_D(s) + \bar{G}_{oN}(s)C_N(s) = \bar{C}_D(s)G_{oD}(s) + \bar{C}_N(s)G_{oN}(s) = I \quad (22.12.22)$$

By using (22.12.22) in (20.5.3) and (20.5.4), the result follows. □□□

From Lemma 22.4, we see again that the closed-loop poles are located at the zeros of $E(s)$ and $F(s)$, as in the SISO case.

22.13 Achieving Integral Action in LQR Synthesis

An important aspect not addressed so far is that optimal control and optimal state-estimate feedback do not automatically introduce integral action. The latter property is an architectural issue that has to be *forced* onto the solution.

We refer the reader to section §18.7, where various ways of introducing integral action (or general IMP) in state-estimate feedback were discussed. For example, from part (b) of section §18.7, we have seen that one way of forcing integral action is to put a set of integrators at the output of the plant.

This can be described in state space form as

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) \\ \dot{z}(t) &= -y(t)\end{aligned}\tag{22.13.1}$$

As before, we can use an observer (or Kalman filter) to estimate x from u and y . Hence, in the sequel we will assume (without further comment) that x and z are directly measured. The composite system can be written in state space form as

$$\dot{x}'(t) = A'x'(t) + B'u(t)\tag{22.13.2}$$

where

$$x' = \begin{bmatrix} x(t) \\ z(t) \end{bmatrix}; \quad A' = \begin{bmatrix} A & 0 \\ -C & 0 \end{bmatrix}; \quad B' = \begin{bmatrix} B \\ 0 \end{bmatrix}\tag{22.13.3}$$

We then determine state feedback (from $x'(t)$) to stabilize the composite system. Of course, a key issue is that the composite system be controllable. A sufficient condition for this is provided in the following.

Lemma 22.5. *The cascaded system (22.13.2)-(22.13.3) is completely controllable, provided that the original system is controllable and that there is no zero of the original system at the origin.¹*

Proof

The result follows immediately from Lemma 20.6 on page 640.

□□□

The final architecture of the control system would then appear as in Figure 22.2.

It is readily seen that the above idea can be extended to any disturbance or reference model, by incorporating the appropriate disturbance-generating polynomials

¹Recall that, if a plant has one or more zeros at the origin, its output cannot be steered to a constant value by a bounded control signal.

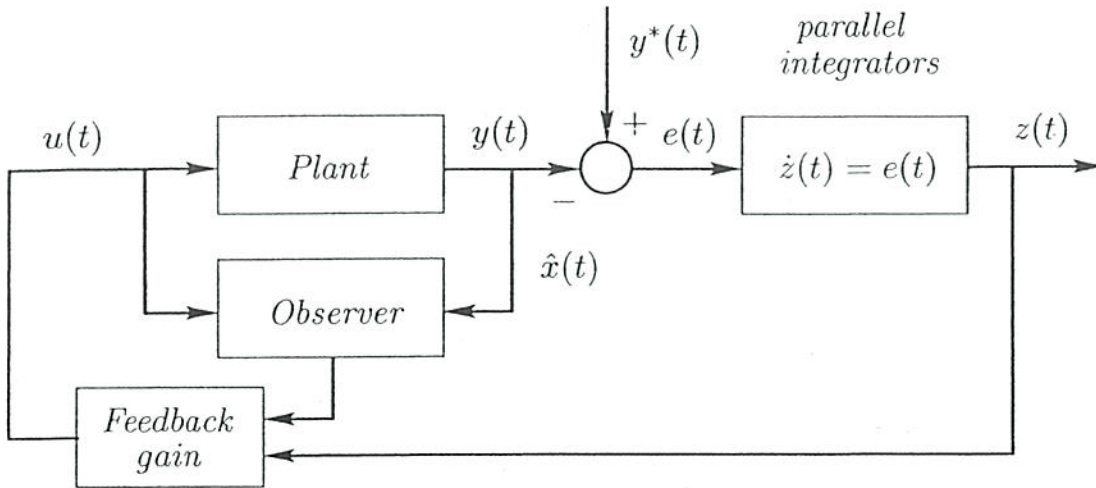


Figure 22.2. Integral action in MIMO control

(Internal Model Principle) in place of the parallel integrators in Figure 22.2. The corresponding version of Lemma 22.5 simply requires that the poles of the disturbance/reference models not be zeros of the original system, exactly as discussed in subsection §20.7.6.

We illustrate the above idea with the following example.

Example 22.4. Consider a system with state space description given by $(A, B, C, 0)$, where

$$A = \begin{bmatrix} -1 & 2 \\ -5 & -3 \end{bmatrix}; \quad B = \begin{bmatrix} -1 & 0 \\ -1 & 2 \end{bmatrix}; \quad C = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix} \quad (22.13.4)$$

Synthesize a linear quadratic optimal regulator that forces integral action and where the cost function is built with Ψ and Φ , where $\Psi = I$ and $\Phi = 0.01 * \text{diag}(0.1, 1)$.

Solution

We first choose the poles for the plant-state observer to be located at -5 and -10 . This leads to

$$J = \begin{bmatrix} -4 & 6 \\ 5 & 2 \end{bmatrix} \quad (22.13.5)$$

Then the regulator will feed the plant input with a linear combination of the plant-state estimates and the integrator states. We choose to build an optimal regulator yielding closed-loop poles located in a disk with center at $s = -3$ and unit radius. (See section §22.8). We thus obtain an optimal feedback gain, given by

$$\mathbf{K} = \begin{bmatrix} -4.7065 & -2.0200 & -8.0631 & 0.0619 \\ -4.8306 & 0.3421 & -8.0217 & -4.0244 \end{bmatrix} \quad (22.13.6)$$

This design is simulated in a loop with the same structure as in Figure 22.2 on the page before. (See SIMULINK file *cint.mdl*.) The tracking performance is illustrated in Figure 22.3, where step references have been applied.

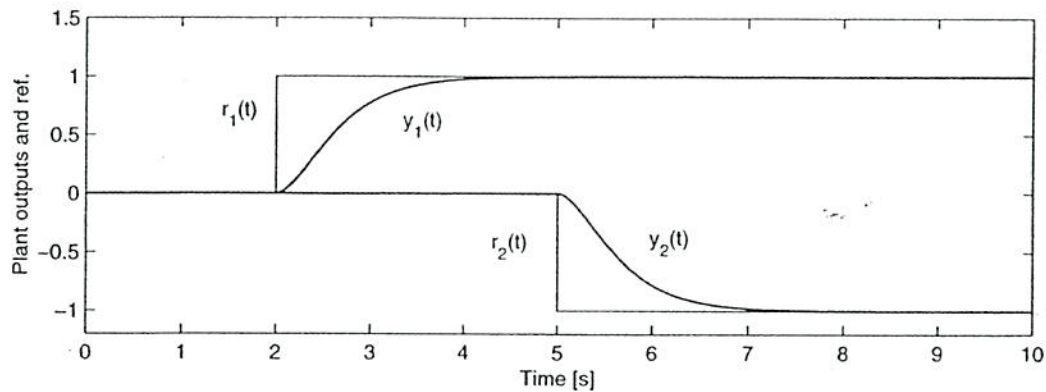


Figure 22.3. Tracking performance of a LQR design with integration

22.14 Industrial Applications

Multivariable design based on LQR theory and the Kalman filter accounts for thousands of real-world applications. Indeed, it was recently pointed out that there are approximately 400 patents per year based around the Kalman filter alone. The authors of this book have also applied these methods in numerous real-world applications. Our belief is that the Kalman filter is one of the most powerful tools available to the control-system designer.

The key issue in using these techniques in practice lies in the problem formulation; once the problem has been properly posed, the solution is usually rather straightforward. Much of the success in applications of this theory depends on the formulation, so we will conclude this chapter with brief descriptions of four real-world applications that the authors have been personally involved with. Our purpose in presenting these applications is not to give full details—that would take us well beyond the scope of this text. Instead, we hope to give sufficient information so that the reader can see the methodology used in applying these methods to different types of problems.

The particular applications that we will review are

- Geostationary satellite tracking
- Zinc Coating-Mass estimation in continuous galvanizing lines

- Roll-Eccentricity estimation in thickness control for rolling mills
- Vibration control in flexible structures

22.14.1 Geostationary Satellite Tracking

It is known that so-called geostationary satellites actually appear to *wobble* in the sky. The period of this wobble is one sidereal day. If one wishes to point a receiving antenna exactly at a satellite so as to maximize the received signal, then it is necessary to track this perceived motion. The required pointing accuracy is typically to within a few hundredths of a degree. The physical set-up is as shown in Figure 22.4.

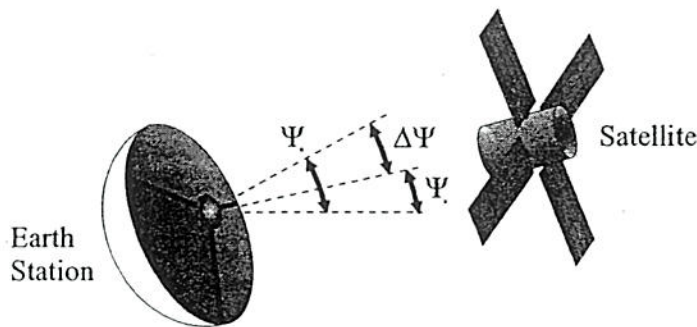


Figure 22.4. Satellite and antenna angle definitions

One could use an open-loop solution to this problem, as follows: Given a model (e.g., a list of pointing angles versus time), the antenna could be pointed in the correct orientation as indicated by position encoders. This technique is used in practice, but it suffers from the following practical issues:

- It requires high absolute accuracy in the position encoders, antenna, and reflector structure.
- It also requires regular *maintenance* to put in new model parameters.
- It cannot compensate for wind, thermal, and other time-varying effects on the antenna and reflector.

This motivates the use of a closed-loop solution. In such a solution, the idea is to move the antenna periodically so as to find the direction of maximum signal strength. However, the data so received are noisy for several reasons, including the following:

- noise in the received signal, p ;
- variations in the signal intensity transmitted from the satellite;

- imprecise knowledge of the beam pattern for the antenna; and
- the effect of wind gusts on the structure and the reflector.

It is a reasonable hypothesis that we can smooth this data by using a Kalman filter. Toward this end, we need first to build a model for the orbit. Now, as seen from the earth, the satellite executes a periodic motion in the two axes of the antenna (azimuth and elevation). Several harmonics are present but the dominant harmonic is the fundamental. This leads to a model of the form

$$y(t) = \Psi_s(t) = x_1 + x_2 \sin \omega t + x_3 \cos \omega t \quad (22.14.1)$$

where $\Psi_s(t)$ is, say, the azimuth angle as a function of time. The frequency ω in this application is known. There are several ways of describing this model in state space form. For example, we could use

$$\frac{d}{dt} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \quad (22.14.2)$$

$$y(t) = C(t)x(t) \quad (22.14.3)$$

$$\text{where} \quad C(t) = [1, \sin \omega t, \cos \omega t] \quad (22.14.4)$$

This system is time-varying (actually periodic). We can then immediately apply the Kalman filter to estimate x_1, x_2 and x_3 from noisy measurements of $y(t)$.

A difficulty with the above formulation is that equation (22.14.2) has zeros on the stability boundary. Hence, inspection of Lemma 22.2 on page 689 indicates that the steady-state filter could experience difficulties. This is resolved by adding a small amount of fictitious noise to the right-hand side of the model (22.14.2) to account for orbit imperfections. Also, note that this is an example of the kind of problem mentioned in section §22.10.6, because the data on satellite location can arrive at discrete times that are nonuniformly spaced.

To finalize the filter design, the following information must be supplied:

- a guess at the current satellite orientation;
- a guess at the covariance of the initial state error ($P(0)$);
- a guess at the measurement-noise intensity (R); and
- a rough value for the added process noise intensity (Q).

A commercial system built around the above principles has been designed and built at the University of Newcastle, Australia. This system is marketed under the trade name ORBTRACK® and has been used in many real-world applications ranging from Australia to Indonesia and Antarctica.

□□□

22.14.2 Zinc Coating-Mass Estimation in Continuous Galvanizing Lines

A diagram of a continuous galvanizing line is shown in Figure 22.5. An interesting feature of this application is that the sheet being galvanized is a meter or so wide and many hundreds of meters long.

The strip passes through a zinc pot (as in the figure). Subsequently, excess zinc is removed by air knives. The strip then moves through a cooling section, and finally the coating mass is measured by a traversing X-ray gauge—see Figure 22.6.

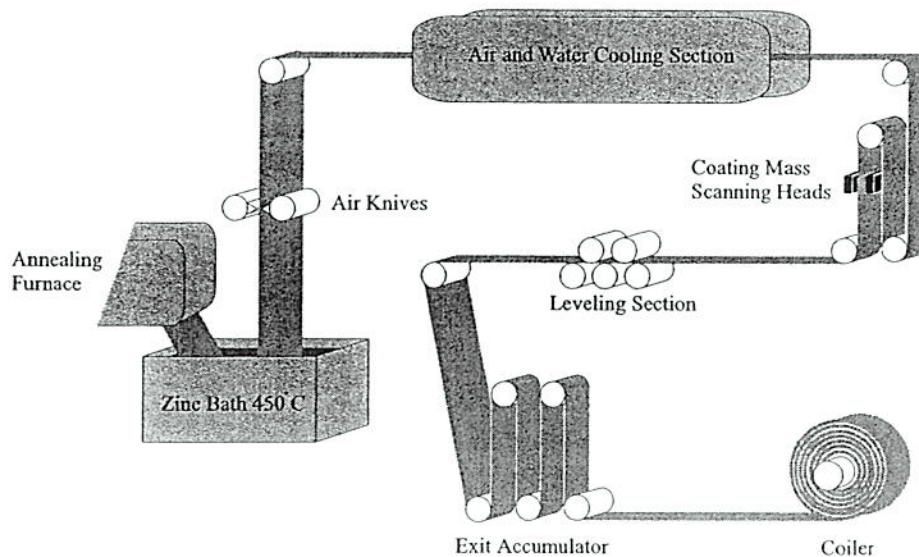


Figure 22.5. Schematic diagram of continuous galvanizing line

If one combines the lateral motion of the X-ray gauge with the longitudinal motion of the strip, then one obtains the zig-zag measurement pattern shown in Figure 22.7.

Because of the sparse measurement pattern, it is highly desirable to smooth and interpolate the coating-mass measurements. The Kalman filter is a possible tool to carry out this data-smoothing function. However, before we can apply this tool, we need a model for the relevant components in the coating-mass distribution. The relevant components include the following.

- *Shape Disturbances* (arising from shape errors in the rolling process)

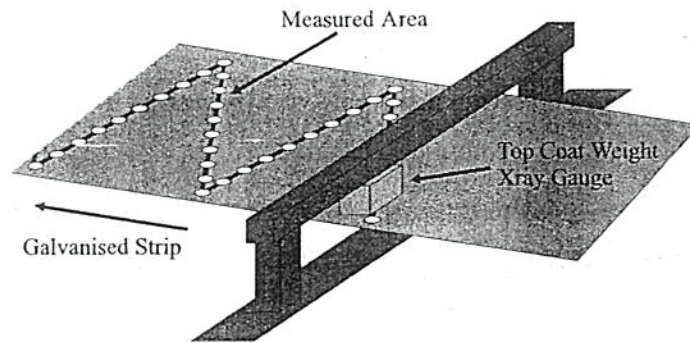


Figure 22.6. Traversing X-ray gauge

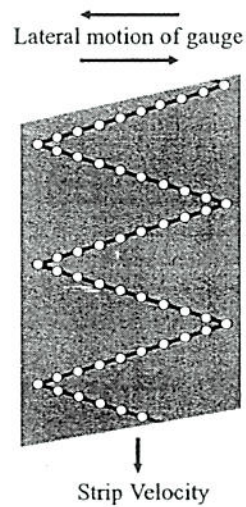


Figure 22.7. Zig-zag measurement pattern

These can be described by band-pass-filtered noise components, by using a model of the form

$$\dot{x}_1 = -\omega_1 x_1 - \left[\frac{\omega_2 \omega_1}{\omega_2 - \omega_1} \right] n \quad (22.14.5)$$

$$\dot{x}_2 = -\omega_2 x_2 - \left[\frac{\omega_2^2}{\omega_1 - \omega_2} \right] n \quad (22.14.6)$$

$$y_{sd} = (1, 1) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad (22.14.7)$$

- *Cross Bow* (a quadratic term arising from nonuniform coating effects)

This is a quadratic function of distance across the strip and is modeled by

$$\dot{x}_3 = 0 \quad (22.14.8)$$

$$y_{cb} = \{d(t)[d(t) - W]\}x_3 \quad (22.14.9)$$

where $d(t)$ denotes the distance from the left edge of the strip and W denotes the total strip width

- *Skew* (due to misalignment of the knife jet)

This is a term that increases linearly with distance from the edge. It can thus be modeled by

$$\dot{x}_4 = 0 \quad (22.14.10)$$

$$y_{sc} = \{d(t)\}x_4 \quad (22.14.11)$$

- *Eccentricity* (due to out-of-round in the rolls)

Say that the strip velocity is v_s and that the roll radius is r . Then this component can be modeled as

$$\dot{x}_5 = 0 \quad (22.14.12)$$

$$\dot{x}_6 = 0 \quad (22.14.13)$$

$$y_e = \left\{ \sin\left(\frac{v_s}{r}\right)t, \cos\left(\frac{v_s}{r}\right)t \right\} \begin{pmatrix} x_5 \\ x_6 \end{pmatrix} \quad (22.14.14)$$

- *Strip Flap* (due to lateral movement of the strip in the vertical section of the galvanizing line)

Let $f(t)$ denote the model for the flap; then this component is modeled by

$$\dot{x}_7 = 0 \quad (22.14.15)$$

$$y_f = \{f(t)\}x_7 \quad (22.14.16)$$

- *Mean Coating Mass* (the mean value of the zinc layer)

This can be simply modeled by

$$\dot{x}_8 = 0 \quad (22.14.17)$$

$$y_m = x_8 \quad (22.14.18)$$

Putting all of the equations together gives us an 8th-order model of the form

$$\dot{x} = Ax + Bn \quad (22.14.19)$$

$$z = y = C(t)x + v \quad (22.14.20)$$

$$A = \begin{bmatrix} -\omega_1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\omega_2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}; \quad B = \begin{bmatrix} -\left(\frac{\omega_2\omega_1}{\omega_2-\omega_1}\right) \\ -\left(\frac{\omega_2^2}{\omega_1-\omega_2}\right) \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (22.14.21)$$

$$C = [1, 1, d(t)[d(t) - W], d(t), \sin\left(\frac{v_s}{d}t\right), \cos\left(\frac{v_s}{d}t\right), f(t), 1] \quad (22.14.22)$$

Given the above model, one can apply the Kalman filter to estimate the coating-thickness model. The resultant model can then be used to interpolate the thickness measurement. Note that here the Kalman filter is actually periodic, reflecting the periodic nature of the X-ray traversing system.

A practical form of this algorithm is part of a commercial system for Coating-Mass Control developed in collaboration with the authors of this book by a company (Industrial Automation Services Pty. Ltd.). The system has proven to be very beneficial in reducing times to accurately estimate coating mass.

22.14.3 Roll-Eccentricity Compensation in Rolling Mills

The reader will recall that rolling-mill thickness-control problems were described in Chapter 8. A schematic of the set-up is shown in the Figure 22.8.

In Figure 22.8, we have used the following symbols:

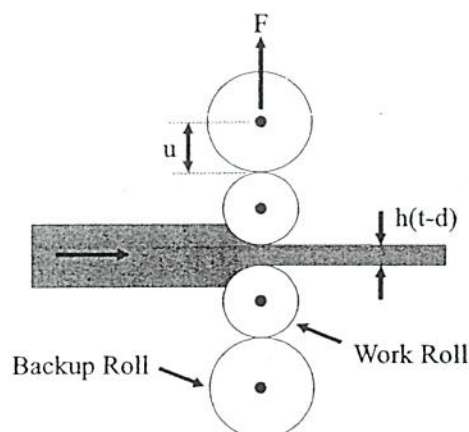


Figure 22.8. Rolling-mill thickness control

- $F(t)$: Force
 $h(t)$: Exit-thickness Measurement
 $u(t)$: Unloaded Roll Gap (the control variable)

In Chapter 8, it was argued that the following virtual sensor (called a BISRA gauge) could be used to estimate the exit thickness and thus eliminate the transport delay from mill to measurement.

$$\hat{h}(t) = \frac{F(t)}{M} + u(t) \quad (22.14.23)$$

However, one difficulty that we have not previously mentioned with this virtual sensor is that the presence of eccentricity in the rolls (see Figure 22.9) significantly affects the results.

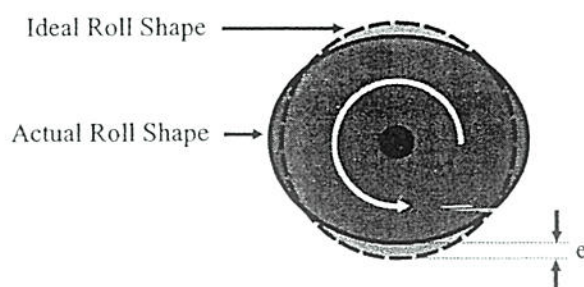


Figure 22.9. Roll eccentricity

To illustrate why this is so, let e denote the roll eccentricity. Then the true roll force is given by

$$F(t) = M(h(t) - u(t) + e(t)) \quad (22.14.24)$$

In this case, the previous estimate of the thickness obtained from the force actually gives

$$\hat{h}(t) = h(t) + e(t) \quad (22.14.25)$$

Thus, $e(t)$ represents an error, or disturbance term, in the virtual sensor output, one due to the effects of eccentricity.

The error $e(t)$ in the virtual sensor output turns out to be very important in this particular application. There is thus strong motivation to try to remove the effect of eccentricity from the estimate $\hat{h}(t)$.

A key property that allows us to make progress on the problem is that $e(t)$ is actually (almost) periodic, because it arises from eccentricity in the four rolls of the mill (two work rolls and two back-up rolls). Also, the roll angular velocities are easily measured in this application by using position encoders. From this data, one can determine a multi-harmonic model for the eccentricity, of the form

$$e(t) = \sum_{k=1}^N \alpha_k \sin \omega_k t + \beta_k \cos \omega_k t \quad (22.14.26)$$

where α_i and β_i are unknown but $\omega_1 \dots \omega_N$ are known.

Moreover, each sinusoidal input can be modeled by a state space model of the form

$$\dot{x}_1^k(t) = \omega_k x_2^k(t) \quad (22.14.27)$$

$$\dot{x}_2^k(t) = -\omega_k x_1^k(t) \quad (22.14.28)$$

Finally, consider any given measurement, say the force $F(t)$. We can think of $F(t)$ as comparing the above eccentricity components buried in noise:

$$y(t) = F(t) = \sum_{k=1}^N x_1^k(t) + x_2^k(t) + n(t) \quad (22.14.29)$$

We can then, in principle, apply the Kalman filter to estimate $\{x_1^k(t), x_2^k(t); k = 1, \dots, N\}$ and hence to correct the measured force measurements for eccentricity.

The final control system is as shown in the Figure 22.10.

An interesting feature of this problem is that there is some practical benefit in using the general time-varying form of the Kalman filter rather than the steady-state filter. The reason is that, in steady state, the filter acts as a narrow band-pass

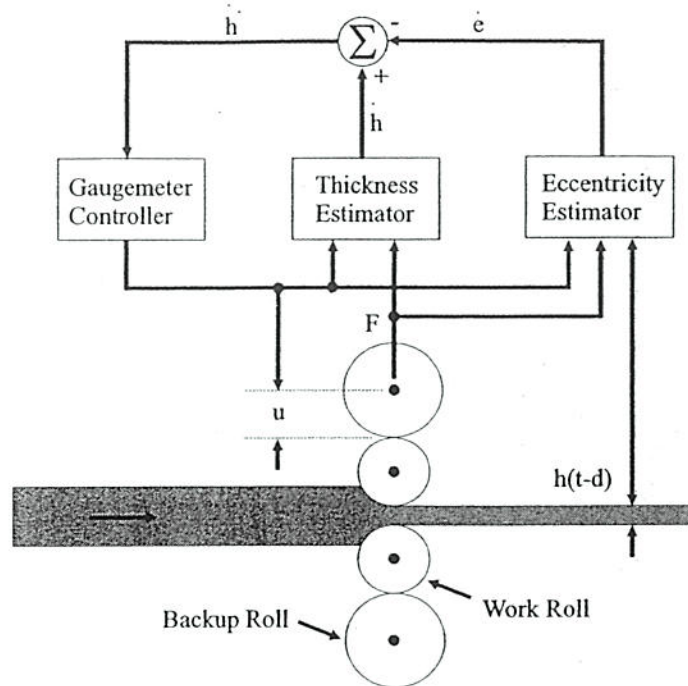


Figure 22.10. Final roll eccentricity compensated control system

filter bank centered on the harmonic frequencies. This is, heuristically, the correct steady-state solution. However, an interesting fact that the reader can readily verify is that the transient response time of a narrow band-pass filter is inversely proportional to the filter bandwidth. This means that, in steady state, one has the following fundamental design trade-off:

- On the one hand, one would like to have a narrow band-pass, to obtain good frequency selectivity and hence good noise rejection.
- On the other hand, one would like to have a wide band-pass, to minimize the initial transient period.

This is an inescapable dichotomy for any time-invariant filter.

Perhaps the astute reader might say: *Well, why not start with a wide-band filter, to minimize the transient, but then narrow the filter band down as the signal is acquired.* If you thought of this, then you are exactly correct. Indeed, this is precisely what the time-varying Kalman filter does. In particular, if one starts the solution of the Riccati equation by setting the initial condition $P(t_o)$ to some large value, then the resultant filter will initially have a wide bandwidth. As the Riccati equation solution proceeds, then $P(t)$ will decrease, along with the associated filter bandwidth. Indeed, the time-varying Kalman filter, gives an *optimal* time-varying trade-off between minimizing the transient time and accurately locking onto the harmonic components.

Thus, the trick to get the best trade-off between filter selectivity and transient response time is to use a time-varying filter.

Another question that the reader may well ask is: *How narrow does the final filter bandwidth ultimately become?* The answer to this question raises another fundamental trade-off in optimal-filter design.

The reader can verify that the model described in $\{(22.14.27), (22.14.28)\}$ has uncontrollable modes (from process noise) on the stability boundary. Careful reading of the properties of the associated filtering Riccati equation given in Appendix D indicates that $P(t)$ converges to a steady-state value in this case but that this happens to be a *zero* matrix. Moreover, the corresponding filter is then not stable. You might well ask: *What has gone wrong?* Well, nothing, really. We told the filter in this ideal formulation that there was no process noise. In this case, the filter could (asymptotically) estimate the states with zero error. Also, once the states were *found*, then they could be predicted into the future by using the resonant model $\{(22.14.27), (22.14.28)\}$. This model has poles on the stability boundary and is thus not stable! (Indeed, the filter bandwidth has been reduced to zero).

The difficulty is that we have over-idealized the problem. In practice, the frequencies in the eccentricity signal will not be exactly known (on account of, e.g., slip). Also, the harmonic components will not be time-invariant (on account of, e.g., roll-heating effects). To model these effects, we can simply add a little process noise to the right-hand side of the model $\{(22.14.27), (22.14.28)\}$. In this case, the model becomes stabilizable *from the noise*. The steady-state solution of the Riccati equation then turns out to be positive definite and the final filter will be stable and of finite bandwidth.

This leads us to another design trade-off. In particular, if one says that the model $\{(22.14.27), (22.14.28)\}$ is poorly known (by setting the process noise intensity Q to a large value), then the final filter bandwidth will be large. On the other hand, if one says that the model $\{(22.14.27), (22.14.28)\}$ has a high degree of validity (by setting the process noise intensity to a small value), then the final filter bandwidth will be small.

In practice, once one has been through such a design as the one above, then it is often possible to capture the key ideas in a simpler fashion. Indeed, this is the case here. In the final implementation, on-line solution of a Riccati equation is typically not used. Instead, a time-varying filter bank is employed that captures the essence of the lessons learnt from the formal *optimal* design. This kind of final *wrap-up* of the problem is typical in applications.

The final system, as described above, has been patented under the name AUSREC[®] and is available as a commercial product from Industrial Automation Services Pty. Ltd.

□□□

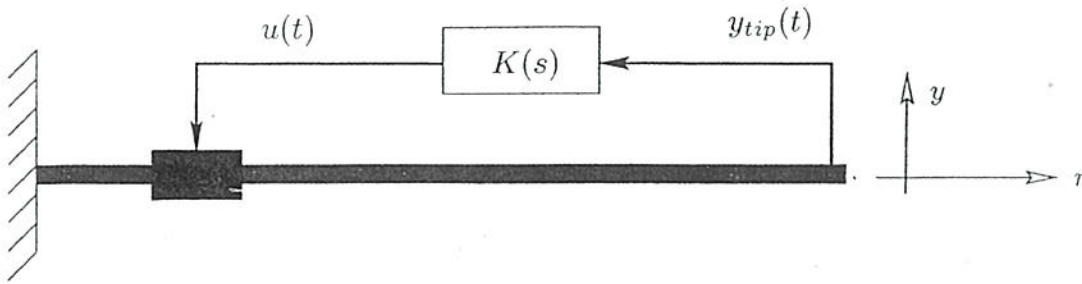


Figure 22.11. Vibration control by using a piezoelectric actuator

22.14.4 Vibration Control in Flexible Structures

As was pointed out in the introduction to this book, new actuators often open up exciting new opportunities for enhanced control. For example, there is substantial interest in new types of electromagnetic actuators and devices that take advantage of the high energy density of smart-material transducer elements. Smart-material-based transducer elements include piezoelectrics, electrostrictions, magnitorestictions, and shape memory alloys.

Many application areas exist for these technologies, especially in the control of lightly decoupled systems.

To illustrate the principles involved in this kind of application, we will consider the problem of controller design for the piezoelectric laminate beam of Figure 22.11. In this system, the measurements are taken by a displacement sensor that is attached to the tip of the beam, and a piezoelectric patch is used as the actuator. The purpose of the controller is to minimize beam vibrations. It is easy to see that this is a regulator problem; hence, a LQG controller can be designed to reduce the unwanted vibrations.

To find the dynamics of structures such as the beam in Figure 22.11, one has to solve a particular partial differential equation that is known as the Bernoulli-Euler beam equation. By using modal analysis techniques, it is possible to show that a transfer function of the beam would consist of an infinite number of very lightly damped second-order resonant terms—that is, the transfer function from the voltage that is applied to the actuator to the displacement of the tip of the beam can be described by

$$G(s) = \sum_{i=1}^{\infty} \frac{\alpha_i}{s^2 + 2\zeta_i\omega_i s + \omega_i^2}. \quad (22.14.30)$$

In general, one is interested in designing a controller only for a particular bandwidth. As a result, it is common practice to truncate (22.14.30) by keeping the first N modes that lie within the bandwidth of interest.

In this example, we consider a particular system described in the references given at the end of the chapter. We consider only the first six modes of this system.

i	α_i	ω_i (rad/sec)
1	9.72×10^{-4}	18.95
2	0.0122	118.76
3	0.0012	332.54
4	-0.0583	651.660
5	-0.0013	1077.2
6	0.1199	1609.2

The transfer function is then

$$G(s) = \sum_{i=1}^6 \frac{\alpha_i}{s^2 + 2\zeta_i\omega_i s + \omega_i^2}. \quad (22.14.31)$$

Here, ζ_i 's are assumed to be 0.002 and α_i 's as are shown in the Table here.

We first design a Linear Quadratic Regulator. Here, the Ψ matrix is chosen to be

$$\Psi = 0.1483 \text{diag}(\omega_1^2, 1, \dots, \omega_6^2, 1).$$

The reason for this choice of state-weighting matrix is that, with this choice of Ψ , the quadratic form $x(t)^T \Psi x(t)$ represents the total energy of the beam at each time corresponding to its first six resonant modes. The control-weighting matrix is also, somewhat arbitrarily, chosen as $\Phi = 10^{-8}$. Next, a Kalman-filter state estimator is designed with $\mathbf{Q} = 0.08I$ and $\mathbf{R} = 0.005$.

To show the closed-loop performance of the controller, we have plotted in Figure 22.12 the open-loop and closed-loop impulse responses of the system. It can be observed that the LQG controller can considerably reduce structural vibrations. In Figure 22.13, we have plotted the open-loop and closed-loop frequency responses of the beam. It can be observed that the LQG controller has significantly damped the first three resonant modes of the structure.

22.15 Summary

- Full multivariable control incorporates the interaction dynamics rigorously and explicitly.
- The fundamental SISO synthesis result that, under mild conditions, the nominal closed-loop poles can be assigned arbitrarily carries over to the MIMO case.
- Equivalence of state-feedback and frequency-domain pole placement by solving the (multivariable) Diophantine Equation carries over as well.

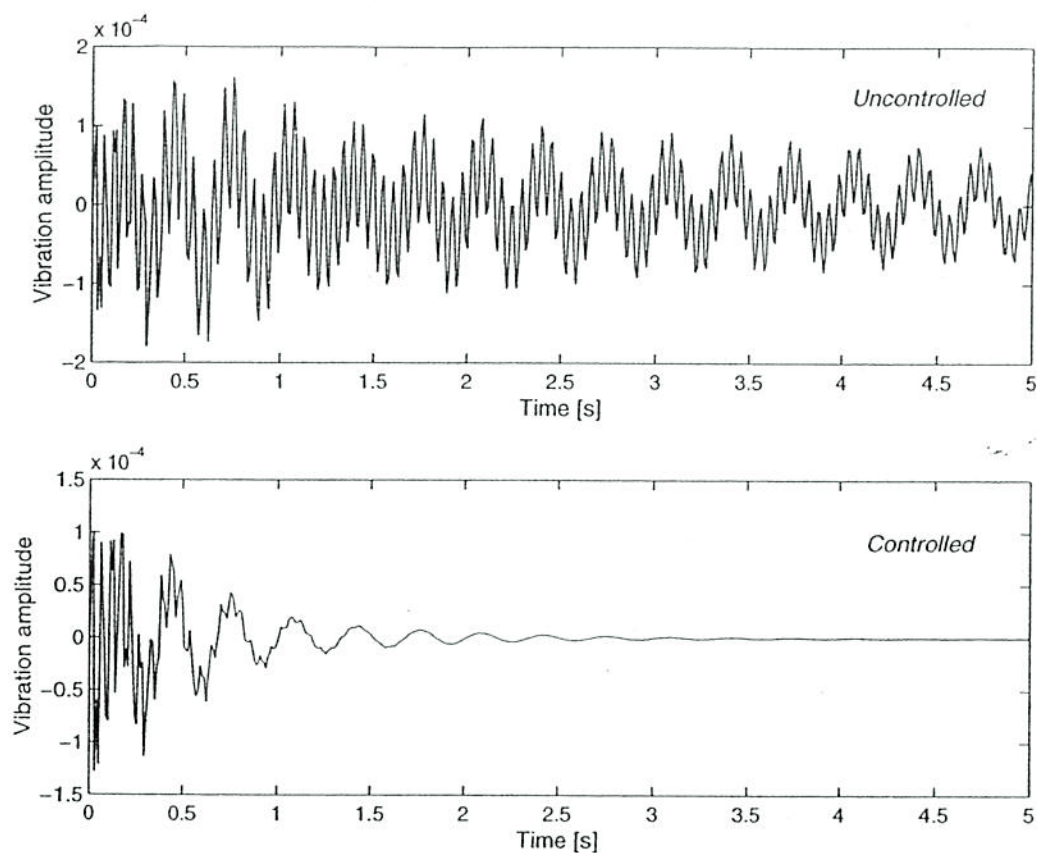


Figure 22.12. Open-loop and closed-loop impulse responses of the beam

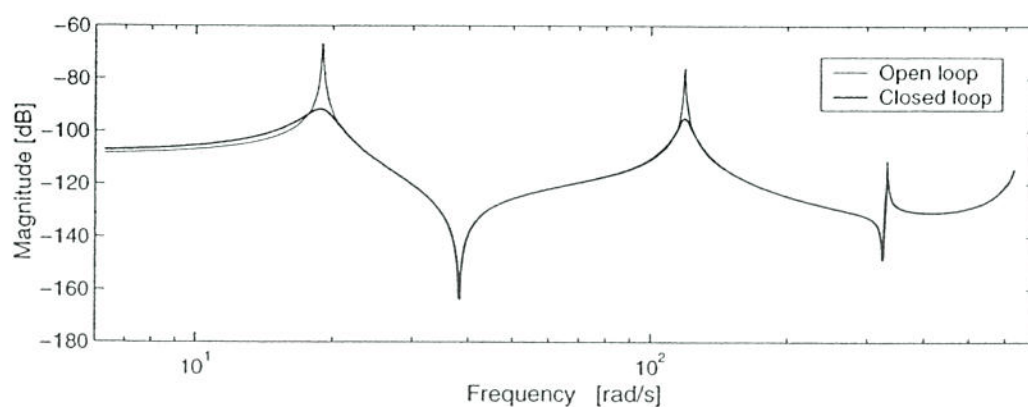


Figure 22.13. Open-loop and closed-loop frequency responses of the beam

- The complexities of multivariable systems, cause criterion-based synthesis (briefly alluded to in the SISO case) to gain additional motivation.
- A popular family of criteria are functionals involving quadratic forms of control error and control effort.
- For a general nonlinear formulation, the optimal solution is characterized by a two-point boundary-value problem.
- In the linear case (the so-called linear quadratic regulator, LQR), the general problem reduces to the solution of the continuous-time dynamic Riccati equation, which can be feasibly solved, leading to time-variable state feedback.
- After initial conditions decay, the optimal time-varying solution converges to a constant state feedback, the so-called steady-state LQR solution.
- It is frequently sufficient to neglect the initial transient of the strict LQR and only implement the steady-state LQR.
- The steady-state LQR is equivalent
 - to a model-matching approach, where a desired complementary sensitivity is specified and a controller is computed that matches it as closely as possible according to some selected measure, and
 - to pole placement, where a closed-loop polynomial is specified and a controller is computed to achieve it.
- Thus, LQR, model matching, and pole placement are mathematically equivalent, although they do offer different tuning parameters.

<i>Equivalent synthesis techniques</i>	<i>Tuning parameters</i>
LQR	relative penalties on control error versus control effort
model matching	closed-loop complementary sensitivity reference model and weighted penalty on the difference to the control loop
pole placement	closed-loop polynomial

- These techniques can be extended to discrete-time systems.
- There is a very close connection to the dual problem of filtering: inferring a state from a related (but not exactly invertible) set of measurements.

- Optimal-filter design based on quadratic criteria leads again to a Riccati equation.
- The filters can be synthesized and interpreted equivalently in a
 - linear quadratic,
 - model-matching, or
 - pole-placement

framework.

- The arguably most famous optimal-filter formulation, the Kalman filter, can be given a stochastic or a deterministic interpretation, depending on taste.
- The LQR does not automatically include integral action; thus, rejection of constant or other polynomial disturbances must be enforced via the Internal Model Principle.

22.16 Further Reading

General optimal control

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